Silent Acceptance: Goodness-of-Fit Alone Cannot Certify Functional Form — Blind Spots of Fit-Based Acceptance and Specification Gates in Physical Model Discovery

Fit quality cannot by itself certify functional form, and the evaluated specification gates share blind spots: 508,825 fitted datasets across six suites plus a 297,000-fit quality re-evaluation — joint-release accounting, budget-audited gate comparison on the verified SVD path, failure registration, and extrapolation evidence.

Anonymous — double-blind review copy

Falsification-side benchmark · goal-switch revision (v4+)
· 2026-09

Table of Contents

Abstract 3

1 Introduction 5

2 Related Work 7

3 Benchmark Design 8

3.1 Design principles . . . 8
3.2 Gates under test . . . 9
3.3 Task suites S1–S3 . . . 10
3.4 Suite S4 — the family-viability battery . . . 10
3.5 Suite S5 — the gate comparison . . . 11
3.6 Suite S6 — real data (World Bank open macro data) . . . 11
3.7 Suite S7 — the joint-silence rate . . . 11

4 Results 12

4.1 Size: false-positive rates on true monomials . . . 12
4.2 Power: raw, then calibrated . . . 13
4.3 Family-stratified results and sensitivity . . . 15
4.4 Noise-layer robustness . . . 15
4.5 Do alarms predict extrapolation failure? . . . 16
4.6 The battery: from rejection to routing . . . 16
4.7 The gate comparison: why not just compare against a nested rival? . . . 18
4.8 Real data: does the map predict live behavior? . . . 19
4.9 Runtime . . . 20
4.10 What the full pipeline actually releases . . . 20
4.11 The joint-silence rate: how often does the whole panel stay quiet? . . . 23

5 Analysis 24

6 Limitations 26

7 Reproducibility statement 28

8 Conclusion 28

References 29

Appendix A — Suite composition 31

Appendix B — Failure registration taxonomy 31

Appendix C — Headline Wilson 95% intervals 32

Appendix D — Result figures 32

Appendix E — The real-engineering data bench (companion to S6) 36

Appendix F — Core code listings 39

Appendix G — Change notes (v2 → v5) 40

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Abstract

Symbolic regression and physical model discovery pipelines certify candidate models with goodness-of-fit metrics — R^2, cross-validation error, information criteria, coefficient standard errors. This paper measures what those metrics can and cannot certify, in both directions, and maps where the classical specification gates share blind spots. The claims are deliberately bounded: fit quality cannot *by itself* certify a functional form — within the single $R^2 \geq 0.99$ stratum, true power laws extrapolate at 0.1–0.2% error while wrong forms reach 100% >10% error on our synthetic bench — a design property of that bench, whose deviations are built to be detectable, not an estimate — and 9.7% of the wrong forms screened on the pool synthesized from real Feynman equations are dangerous; and the evaluated specification tests (RESET and variants, nested-rival comparison) remain jointly blind to some form mismatches — wrong forms passing an $R^2 \geq 0.99$ screen are jointly missed by all four gates at 0.67% overall, and among *dangerous* wrong forms (extrapolation error >10%) the joint release rate is 6.9%. With that screen removed the same four gates are jointly silent on **37.8%** of wrong forms drawn from real equations at n = 50 — roughly two orders of magnitude above the synthetic bench's 0.15–0.22% on the same unscreened basis — and adding selection-side gates raises rather than lowers the calibrated figure (38.6% → 42.2%, against 90.5% silence on true forms; a silent certificate therefore carries only a 2.2× likelihood ratio). We do not claim, and our data contradicts, stronger readings: fit screening combined with the gates carries real information (a full-green certificate is 65× less likely for a dangerous wrong form than for a true one: .013 against .845); the failure is non-certification and non-detection of specific mismatch directions, not statistical independence.

On the synthetic bench, within the single $R^2 \geq 0.99$ stratum, true power laws extrapolate at ≈0.1–0.2% error while wrong forms sit at 100% >10% error: the same certificate, opposite function truth. The paper's headline quantity is the *joint* event on the same models — a dangerous wrong form (extrapolation error >10%) that passes the $R^2 \geq 0.99$ screen AND is jointly missed by all four gates: 0.67% of screened wrong forms (481/71,992), but 6.9% of dangerous ones; and the release rate of dangerous forms falls from 7.8% (n = 50) to 0.2% (n = 1000) counting all dangerous fits, and from 13.2% to 1.2% counting screened dangerous models — a fall driven by the fit screen tightening at scale, not by any single gate (Section 4.10b). A full-green certificate carries real information (a dangerous wrong form is 65× less likely to hold one than a true form: .013 against .845); what it cannot do is *exclude* the failure mode.

We introduce the falsification-side benchmark that quantifies this. Six suites total 1,480 grid cells and 508,825 fitted datasets, with 472,500 further held-out evaluations and a quality re-evaluation over 297,000 of these fits: (S1) 19 controlled synthetic scenarios; (S2) 32 tasks from the SRSD-Feynman registry under official sampling configurations; (S3) 30 Feynman equations stratified by structural family; (S4) a 188-cell family-viability battery extending the same machinery from rejection to routing; (S5) a gate comparison placing the benchmark's best test against the "obvious alternative" it had never faced — a nested quadratic rival adjudicated by 5-fold cross-validation or AICc; and (S6) six macroeconomic

relations from World Bank open data across four years and ~190 economies per cell. A seventh suite (S7) re-samples S2's 18 non-monomial equations on an independent seed grid — 108,000 further fits — to measure the panel's joint-silence rate with the fit screen removed; its counts are reported separately and are not folded into the 508,825 above. All methods run under one protocol: a common significance level; power compared only at matched empirical false-positive rate; preregistered labels; explicit registration of every degenerate or rank-deficient evaluation.

Five findings stand out. **First**, the phenomenon is real and silent: a power law fitted to Prandtl–Karman friction data certifies itself at $R^2 = 0.99$ through every metric a pipeline computes, while extrapolating at −20% one decade out. **Second**, the benchmark caught — and its revision retracted — its own most striking claim: our harness's HC3-Wald path rejects at rate 1.00 on 5 of 13 physics-derived tasks whose built designs are near-collinear (pooled 42–56% false alarms, flat in n), but an independent SVD-based implementation (statsmodels) is nominal-compatible (.067, Wilson [.034,.126]) on the same data and agrees with ours to three digits elsewhere: the failure is a silent numerical pathology of pinv-squared-conditioning inference paths, not of the robustness correction — a deployment hazard discovery code should test for. **Third**, no test is uniformly best; rankings invert under matched false-positive rates and again across task families, and each method's strength sits on another's blind spot. **Fourth**, the battery extends rejection to routing after its null is rebuilt per family — null behavior is coordinate-dependent (the residual permutation is empirically calibrated in log-log coordinates on every pool we measured; 30–34% false-escalation in rational coordinates until a wild bootstrap repairs it). **Fifth** — the benchmark's most uncomfortable results — the nested quadratic rival is the better default detector wherever a reference pool exists, and the naive OR union of the permutation gate with the CV rival, previously the best-recall cell, owes that advantage to a larger joint false-positive budget: recalibrated to a 5% joint budget, the union matches but does not beat AICc-quad. On real macroeconomic data the gates agree where signal is clear, diverge where the map predicts weakness, and show an observational pattern consistent with the synthetic dilution signature; S6's six relations carry no ground truth and license signature-consistency claims only, not rates. The deployment ordering: default AICc-quad when a reference pool is available; the permutation gate when a self-calibrating p-value is required; recall-critical deployments should budget for AICc-quad, not for an uncalibrated union. We release all generators, per-cell results, failure registrations, and content-hashed configurations under fixed seed rules.

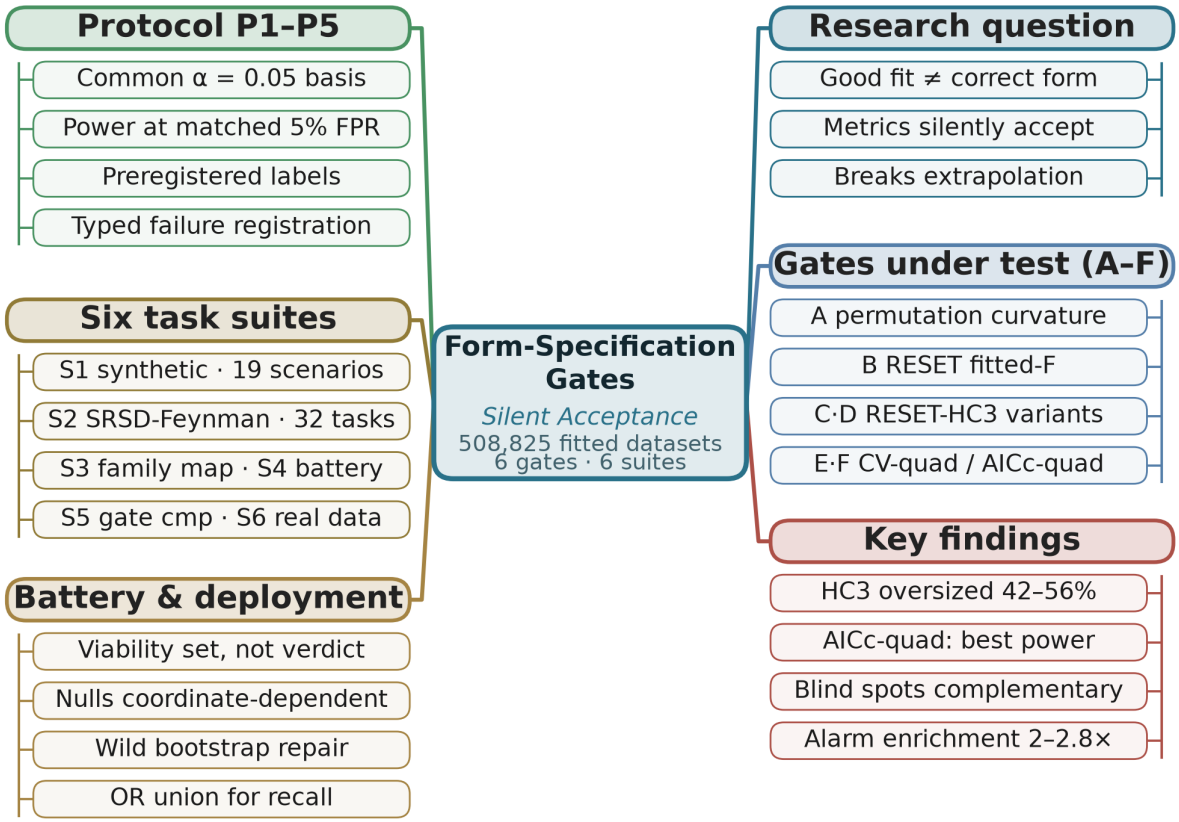


Figure 1 — Mind map of the benchmark paper: research question, the gates under test, key findings, protocol, suites, and the family battery.

1 Introduction

A regression model can be simultaneously excellent on every certification metric a discovery pipeline computes and wrong in its functional form. A power law fitted to Prandtl–Karman friction data reaches R^2 = 0.99, passes k-fold cross-validation, wins its information-criterion comparison, and reports small standard errors — while extrapolating with −20% error one decade outside the training domain. This is not hypothetical: the failure is reproduced as a controlled scenario in our laboratory notebook and appears in this benchmark as scenario S1 (Section 3.3), where all four specification tests detect it at power ≥ 0.99. Fit quality and form correctness are different axes, and current discovery practice measures only the first.

This paper sharpens that observation into falsifiable, bounded claims and measures both directions. Claim one — fit quality cannot by itself certify a functional form. On our synthetic bench (Section 4.10), within the single $R^2 \geq 0.99$ stratum, true power laws extrapolate at 0.1–0.2% error while wrong forms sit at 100% >10% error: the same certificate, opposite function truth. Claim two — the evaluated specification gates remain jointly blind to some form mismatches. The joint event on the same models (Section 4.10): a dangerous wrong form (extrapolation error >10%) that passes the $R^2 \geq 0.99$ screen and is missed by all four gates occurs for 0.67% of screened wrong forms but 6.9% of dangerous ones; the extreme case, equation ii.11.28, is jointly missed at 83–84% (n = 50/200) despite fitting at $R^2 \approx 0.002$. Both numbers are necessary-negation results, one clean counterexample each would establish the logical point; what 297,000 re-evaluated fits add is the *extent* — where the blind spots sit (odd-symmetric and interaction

mismatch directions), how often they fire, and how they scale with sample size. We do not claim the absence of a statistical relationship: fit screening plus the gates carries real information (a dangerous wrong form is 65× less likely to earn a full-green certificate than a true one), and the alignment diagnostic corr($\hat{y}^2$, residual) predicts gate behavior at Spearman +0.47 to +0.76. The failure mode is specific: no combination of these tools *excludes* being wrong in a direction they do not span — certificates of correctness cannot be built from fit magnitude alone.

The statistics literature has owned the form axis for over fifty years. Ramsey's RESET [1] tests whether powers of the fitted values carry additional signal; built-regressor variants use powers of the regressors; White [2] and MacKinnon–White [3] provide heteroskedasticity-robust inference. And the workflow alternative that requires no new statistics at all — fit a nested rival family and let cross-validation or an information criterion arbitrate — has been standard practice in applied regression for as long. Yet none of this is measured by symbolic regression benchmarking. SRBench [5] measures whether discovery methods find the right expression across 252 problems; SRSD [6] redesigns Feynman-derived tasks with realistic sampling ranges. In neither framework does anything measure whether an attached falsification layer would reject a wrong form. The asymmetry has practical consequences: a discovery pipeline with no working falsification layer silently accepts misfit forms, and its downstream use — extrapolation, engineering design, scientific claim-making — inherits the error silently.

Measuring the falsification side faithfully requires resolving a protocol problem that, to our knowledge, no prior work addresses for this test family: the candidate gates reject on incommensurable bases. A bootstrap-t threshold ("alarm when $|t| > 3$"), an asymptotic F p-value ("reject when $p < 0.05$"), and a cross-validated RMSE difference ("escalate when the rival wins") measure different things; comparing them directly measures threshold placement, not detection. Our first pilot compared a native $|t| > 3$ alarm against $p < 0.05$ and produced inverted rankings that vanished under a common basis. The protocol below (Section 3.1) rests on three commitments: one decision basis per comparison; power compared only at a matched empirical false-positive rate estimated on true-monomial tasks; and explicit registration of every failure.

This paper asks its benchmark the three questions a reviewer would ask, in the order they would ask them. (i) *Do the classical tests work at all on physical data distributions?* — Sections 4.1–4.5: calibration, power, blind zones, extrapolation semantics. (ii) *A rejection terminates inquiry — then what?* — Section 4.6: a family-viability battery that routes instead of refusing, once its null is rebuilt per coordinates. (iii) *Why not just fit a nested rival and compare?* — Section 4.7: because until it is measured, that question has no answer; measured, it wins, and the benchmark says so. A real-data suite (Section 4.8) then asks the external-validity question: does the synthetic failure-mode map predict anything about live observational data? It does, including one live reproduction of a synthetic blind spot.

The question this benchmark answers is narrow and falsifiable: **when the ground truth is known and the candidate form is wrong, which gates detect the misspecification — at what false-alarm cost, on what reference budget, and do the synthetic failure modes transfer to real data?** We contribute: (i) the benchmark — six core suites, 508,825 fitted datasets plus 472,500 held-out evaluations, released generators with seed rules making any cell independently reproducible; (ii) the protocol — matched-basis comparison with empirical-FPR calibration and failure registration; (iii) the failure-mode map — including a silent numerical failure of pinv-based HC3-Wald inference on near-collinear physics-derived designs (independently verified; the correction itself is not implicated), the near-blindness of fitted-F

RESET to rational laws, and the measured dominance and calibration cost of the nested-rival gates; (iv) the family-viability battery — 94,000 datasets showing rejection extends to routing once — and only once — the null is rebuilt per coordinates; and (v) a real-data suite showing the map's mechanisms operating outside the laboratory; and (vi) the joint-release analysis — Section 4.10, which connects the fit screen, the four gates, and extrapolation risk on the same models: what the full pipeline releases, how dangerous the releases are, and how the release channel scales with sample size. What we deliberately do not contribute is a winning detector; the benchmark's own results (Section 4.7) demote its best test from default gate to portfolio member, and the paper is organized around that honesty.

2 Related Work

Specification testing. RESET [1] augments the regression with powers of the fitted values and tests the joint significance of the added coefficients; the exog variant uses powers of the regressors instead. Heteroskedasticity-consistent covariance estimation [2] and its HC3 leverage-weighted refinement [3, 4] are the standard small-sample corrections and are the default in several mainstream econometrics and statistics packages. These results are derived for linear models under fixed or quasi-fixed designs; none of the classical treatments contemplates log-transformed physical laws whose regressors span only a few orders of magnitude — the regime where our harness's own pinv-based inference path silently failed (Section 5, Q3) and where the verified SVD implementation of the same tests behaves nominally (Section 4.1).

Model selection as a specification gate. Comparing a null model against a nested rival by information criterion [12] or cross-validation [13] is the oldest workflow-level misspecification check in applied practice. It is so obvious that a reviewer of any new test will ask for it; it is so workflow-dependent (which rival? which criterion? which threshold?) that benchmarking it requires exactly the protocol this paper introduces — a frozen rival family, frozen decision statistics, and thresholds calibrated on a reference pool. Suite S5 is, to our knowledge, the first head-to-head we know of between a resampling-null specification test and nested-rival selection gates under matched false-positive rates.

Resampling nulls under non-i.i.d. errors. The wild bootstrap [10, 11] was developed for inference under heteroskedasticity, and its permutation cousin requires exchangeability. The battery's mechanism finding (Section 4.6) is a compact new instance of this classical dichotomy: a log transform homoskedasticizes multiplicative noise, so residual permutation is valid in coordinates ending in log y and invalid in coordinates ending in y or 1/y — the null's validity inherits the noise geometry of the coordinates the test runs in.

Symbolic regression benchmarks. SRBench [5] established the discovery-side evaluation: 14 SR methods and 7 ML baselines on 252 problems, scored by exact recovery and held-out accuracy. SRSD [6] reconceived the task distribution itself — realistic per-variable ranges reviewed against the physics — and is the source of our task configurations. Both measure the discovery side; a pipeline that returns a wrong expression and one that flags it score identically. Our generators deliberately reuse SRSD's official sampling objects so that the falsification-side pools are the same distributions the discovery side already standardizes on.

Physics-informed and dimensionally-aware discovery. AI Feynman [7] uses physical units to constrain the hypothesis space; measurement-driven discovery distills governing equations and structure directly from data [8]; PySR [9] represents the modern search side. Dimensional correctness is necessary but not sufficient for form correctness: every ground-truth law in our suites is dimensionally self-consistent, yet each non-monomial law is a form misspecification for a power-law model. The falsification side complements these constraints — units narrow what to search; specification gates check what was found.

Honest-failure evaluation. Existing SR benchmarks report no analogue of a false-positive rate because they contain no explicit rejection mechanism to calibrate. We are not aware of a symbolic regression benchmark that measures attached falsification behavior, per family, with preregistered labels and failure registration — or one that benchmarks its own proposed test against the nested-rival alternative its reviewers would name. This is the gap the present benchmark occupies.

The wider specification-testing literature. Econometrics and statistics hold many more form-specification tools than the gates tested here: smooth residual tests, generalized likelihood-ratio and conditional moment tests, rainbow and related one-degree-of-fitting checks, spline/GAM residual comparison, kernel-based lack-of-fit statistics, cross-fitted and sample-splitting variants, and conformal-style selective-risk diagnostics. S5's nested-quadratic rival is the discovery community's canonical objection, not a survey of this literature; LOOCV, BIC, held-out likelihood, or non-nested rivals may rank differently. The gates were chosen because they are what discovery pipelines actually attach today (RESET variants, robust inference) plus the workflow alternative a reviewer would name (nested-rival selection); the harness is gate-agnostic — a new gate is one statistic-plus-decision function over the same fitted cells — and extending the benchmark to these tests is future work, not a closed set.

3 Benchmark Design

3.1 Design principles

P1 — Common rejection basis. Within any comparison, all gates decide on the same basis at the same significance level $\alpha = 0.05$. The permutation curvature test's empirical p-value is computed from $B = 999$ permutations of the residual-permutation null; RESET variants use asymptotic F reference distributions; S5's selection gates use thresholded statistics, with thresholds at naive zero and at an H0-pool-calibrated value — and naive versus calibrated results are reported side by side, because raw tables are what current practice would see.

P2 — Power only at matched empirical FPR. Headline power comparisons calibrate each method's decision threshold to a 5% empirical false-positive rate, estimated on pooled true-monomial tasks within the same suite, sample-size stratum, and noise regime.

P3 — Preregistered labels. H0 (strict monomial: the log-log mean function is exactly linear) versus H1 (any non-monomial structure) is fixed from symbolic structure of the ground truth before any test runs. The family stratification of S3, the battery's truth registry, the S5 gate definitions, and the S6 relation list are likewise frozen in code before their runs; no label or configuration is revised after seeing results.

P4 — Explicit failure registration. Every evaluation that cannot produce a valid decision is registered with a typed reason — zero-residual degeneracy, design rank deficiency, insufficient residual degrees of freedom, leverage degeneracy, singular covariance, statistic mismatch, generation failure (S5 SRSD side), indicator archival (S6) — and excluded from rate numerators while remaining in the denominators' audit trail. The zero-noise grid cells exist to exercise this path.

P5 — Official data configurations. S2/S3 tasks reuse the official per-variable sampling objects of the source registry; tasks whose inputs or outputs are not strictly positive under the official configuration are marked not-applicable and excluded — we do not shift or rescale data to make them log-fittable, since that changes the hypothesis under test. S6 uses a public observational source (World Bank Open Data API [14]) with the relation list frozen before any gate ran; the United Nations Foundation itself publishes no dataset API, and the World Bank is the UN-system institution with the open macro data of record.

3.2 Gates under test

All gates operate in log-log space on positive data: let $ly = \log y$ and $X = [1, lx_1, \ldots, lx_k]$ be the design of the candidate power-law model.

A. Permutation curvature (form_check). Adds a single mean-centered quadratic column $q = \text{mean}_j((lx_j - \bar{lx}_j)^2)$ and takes the added coefficient c_2 as statistic; size evidence comes from B = 999 residual permutations (vectorized through the pseudo-inverse row), with a statistic-mismatch sentinel comparing wrapper and native values on every call. Validity note: permuting *estimated* residuals assumes they are exchangeable, which OLS residuals are not exactly — the permutation null here is an approximation whose quality is an empirical question, not a finite-sample exactness theorem. What the benchmark establishes is calibration of this construction on our H0 pools (Tables 1–2); log-transformed multiplicative noise and favorable designs plausibly help, and §6 lists the stress tests (leverage-adjusted residuals, wild and block nulls, cluster dependence, fixed-design variants) that would bound where the approximation breaks.

B. RESET fitted, classical F. Adds $\hat{y}^2$ and $\hat{y}^3$; classical F test of joint significance.

C. RESET fitted, HC3-Wald. Same augmented design; HC3 sandwich Wald statistic on the added coefficients, referred to $F(q, n - k)$.

D. RESET exog, HC3-Wald. Built regressors are powers of each log-regressor; HC3-Wald as in C.

E. CV-quad (S5). The "obvious alternative": rival = full quadratic expansion of the log features (linear terms, per-feature squares, pairwise cross-terms; minimum-norm least squares absorbs rank deficiency). Statistic $D = \text{RMSE_cv(power)} - \text{RMSE_cv(quad)}$ over 5 deterministic folds; escalate if $D > \delta$. Naive $\delta = 0$; calibrated δ^* chosen on the H0 pool to FPR = 5%.

F. AICc-quad (S5). Same rival; escalate if $\text{AICc(power)} - \text{AICc(quad)} > m$, k = column rank. Naive $m = 0$; calibrated m^* as in E.

Configurations were frozen before all main runs — powers (2, 3); B = 999; $\alpha = 0.05$; the rival family and criteria of E/F; the grids below — and are identical across suites. The same harness invokes all gates on every dataset, so comparisons share data, seeds, and failure handling exactly.

3.3 Task suites S1–S3

S1 — Synthetic control (19 scenarios × 500 seeds). Three monomial H0 scenarios (univariate; two-variable; two-variable with $\rho \approx 0.9$ correlated design). Eleven structural H1 scenarios: quadratic curvature ($c = 0.10$); cubic (standardized odd term); multiplicative interaction (log-space cross term); threshold (exponent $0.4 \rightarrow 1.0$ at the domain midpoint); exponential law; logarithmic law; rational (saturating) law; omitted variable; dilution (one of two variables curved); opposing curvature (+0.12, −0.12 on an exchangeable design, so the mean-square column's population coefficient is exactly zero); collinear-quadratic ($\rho \approx 0.95$ design with curvature). Five noise-layer H1 scenarios have strict-monomial mean structure but non-Gaussian noise — Student-t(3); SD growing with the log prediction (0.3× to 1.7×); 2% point outliers (+8σ); 6% group outliers (+5σ); AR(1) $\rho = 0.6$ — rejections there are false alarms by construction and are never pooled into structural power. Grid: $n \in \{50, 200, 1000\}$; relative Gaussian noise $\sigma \in \{0, 0.5\%, 1\%, 3\%, 5\%\}$ in log space; 500 seeds per cell via a seed rule embedding the grid position.

S2 — SRSD-Feynman tasks (32 tasks × 500 seeds). From the 120-equation registry, 32 tasks pass the positivity screen under official configurations: 13 strict-monomial H0 and 19 non-monomial H1. Each training fit ($n \in \{50, 200, 1000\}$) is accompanied by a 2,000-sample same-distribution test set and a far-field probe: the first regressor extrapolated ×3 beyond both training domain edges, scored against the closed-form truth in a numerically stable form.

S3 — Family-stratified Feynman equations (30 equations × 30 seeds). Structural family classification from symbolic form: exact power law (12 H0); fractional power (1 H0); exponential (3 H1); rational (6 H1); trigonometric (5 H1); plus not-applicable members with explicit reasons. Each H1 equation runs over relative noise {0.5%, 1%, 3%, 5%} × {homoskedastic, heteroskedastic} × sampling-range scale {official, ×0.5, ×2}; H0 equations run the noise grid at official range.

3.4 Suite S4 — the family-viability battery

Each test of §3.2 evaluates a single null form; a rejection terminates inquiry. The battery registers four families with their linearizing coordinate pairs — power (log x, log y); exponential (x, log y); logarithmic (log x, y); rational (1/x, 1/y) — and for each family F runs A's construction in F's coordinates, returning the viability set $V = \{F : p_F \geq \alpha\}$ rather than a verdict. Columns are standardized before squaring so features at different scales weight equally, and the null is a wild bootstrap (Rademacher signs; $B = 999$) rather than a permutation — for the reason §4.6 documents: **null validity is a coordinate-dependent property**. The prediction (preregistered before any run) was that permutation, valid wherever the v-coordinate is log y, would break in coordinates ending in y or 1/y; the probe measured exactly that, and the formal run uses the wild null throughout. Three errors are reported separately and never merged: false exclusion (truth absent from V), false inclusion (wrong family present), and escalation on structures no family owns — the correct outcome, never scored as a win. Grid: $n = 200$; rel $\in \{0.5–5\%\}$; 500 seeds; 188 cells; 94,000 datasets, all 19 S1 scenarios with truth-family labels frozen in code plus 14 preregistered SRSD H1 equations × {homoskedastic, heteroskedastic}.

3.5 Suite S5 — the gate comparison

S5 exists because of an objection this benchmark could not answer from its own data: *why not skip the test statistic entirely and let a nested rival arbitrate?* Gates E and F (§3.2) are that objection, made runnable. Design, frozen in `gate_compare.py`'s docstring before the run: benches = S1's 19 scenarios and S2's 32 tasks; grid $n \in \{50, 200\}$, rel = 3%, 200 seeds per cell (seed bases 300,000 / 400,000 — disjoint from all prior suites); rival = full quadratic expansion; E and F thresholds naive and H0-pool-calibrated per bench × n; A run under its standard protocol on identical data. Metrics: H0 size (naive and calibrated), structural-H1 detection at calibrated FPR 5%, noise-layer fire rates (should stay quiet), per-scenario and per-equation detail (never macro-averaged away), and cost in fit units. One asymmetry is inherent and is reported rather than hidden: A's permutation null is self-calibrating, while E/F at naive thresholds inherit nested-comparison tie bias; the calibrated variants price what a reference pool buys.

3.6 Suite S6 — real data (World Bank open macro data)

Synthetic and equation-derived suites have ground truth by construction; the deployment question is whether the synthetic failure-mode map predicts anything on observational data, where no truth exists. S6 is deliberately small, preregistered, and divergence-oriented: six macroeconomic relations (target, features) × four years (2019, 2021, 2022, 2023), cross-country cross-sections of 184–199 sovereign economies (World Bank aggregates excluded), from eight indicators pulled from the World Bank API with raw responses cached for reproducibility — life expectancy ~ GDP per capita (PPP); infant mortality ~ GDP per capita; GDP ~ population; CO_2 per capita ~ GDP per capita; internet penetration ~ GDP per capita; and life expectancy ~ GDP per capita + health expenditure per capita (two-feature). Gates A–D run at 5 permutation seeds per cell. These seeds are computational replicates of one p-value estimator on one fixed dataset — not five independent samples — so they are used only to report Monte Carlo stability (the empirical p has resolution 1/1000 at B = 999; across the 24 S6 cells the median p spread is .009 and the max .094), never as a sample-size or stability claim; a preregistered probe adds three agricultural and energy relations over three years, including a built-in control-contrast pair — cereal yield ~ fertilizer (one feature) versus cereal yield ~ fertilizer + energy use per capita (two features) — chosen to test, on live data, whether the dilution mechanism of S1 (a second correlated curved feature halving the mean-square column's signal) operates outside the laboratory. No calibration or power claim is made on S6; its contract is external validity of *mechanisms*, not of rates. An engineering companion bench (Appendix E) extends the same contract to physical-engineering data: nine open sources, 91 datasets, 150 preregistered relations, run as a single-realization smoke pass.

3.7 Suite S7 — the joint-silence rate

Section 4.10 asks what the *pipeline* releases (fit screen, then gates). S7 asks a different question: with the fit screen removed,

how often does the entire gate panel stay silent on a wrong form? The two must never be pooled — S7's denominator is every evaluated H1 fit, not the screened subset — and they price different deployment questions. Section 4.10 prices the pipeline a practitioner would actually run; S7 prices the gates themselves.

Design. The 18 non-monomial H1 equations of S2 that survive the far-field probe (`feynman-i.30.5` withdrew, v5 App. B). Per task and n ∈ {50, 200, 1000}, 2,000 datasets — 36,000 per n, on a seed grid disjoint from S2's 500-seed draw. Gates A/B at native thresholds, C/D on the SVD path. Two threshold regimes: naive α = .05, and H0-pool calibrated to the same per-gate 5% budget. Joint silence requires that **no gate rejects and no gate declines**: a `fail_*` registration is an honest refusal of the evaluation, not a certificate, and counting it as silence would flatter the panel.

A second arm adds the model-comparison gates (CV-quad, AICc-quad — the S5 members) on `gate_cmp`'s SRSD arm (3,800 datasets; a subset of the 20,400-cell S5 bench, not the whole of it) to test whether selection-side gates seal the leaks the specification gates leave open. Its H0 reference is that same panel's silence rate on true power laws. Because the arms rest on different benches, **their levels are not comparable across arms**; only within-arm contrasts are used below.

Conditional cost is measured separately: among jointly-silent wrong forms, the far-field (one-decade) extrapolation error distribution. A silent certificate whose median far-field error is small is not thereby safe; the claim is about the tail.

4 Results

Table numbering is consecutive: Tables 1–2 size, Tables 3–4 power, Table 5 the family map, Table 6 noise, Table 7 extrapolation, Tables 8–10 the battery, Tables 11–12 the gate comparison, Tables 13–14 real data, Table 15 the no-entailment summary, Table 16 the joint-silence rate. All rates are rejection fractions; Wilson 95% intervals for headline rates appear in Appendix C. Seed counts per cell: 500 (S1, S2, S4), 30 (S3), 200 (S5), 5 (S6 — permutation-p stability, not rate estimation), 2,000 per task (S7; equations, not datasets, are the independent unit there — see §6).

4.1 Size: false-positive rates on true monomials

Table 1 — S1 synthetic H0, false-positive rate at α = 0.05 (Gaussian noise pooled over σ ∈ {0.5%…5%}; 4,500 fits per cell).

Method	n=50	n=200	n=1000
A — permutation curvature	.047 [.041–.053]	.037 [.032–.043]	.041 [.036–.047]
B — RESET fitted F	.045 [.039–.051]	.041 [.036–.047]	.038 [.033–.044]
C — RESET fitted HC3	.048 [.042–.054]	.045 [.039–.051]	.038 [.033–.044]
D — RESET exog HC3	.054 [.048–.061]	.042 [.036–.048]	.040 [.034–.046]

Table 2 — S2 physics-derived monomial tasks (13 SRSD-Feynman equations; synthetic draws from real equations, not measurements), false-positive rate at α = 0.05 (26,000 fits per cell; C/D on the verified SVD path).

Method	n=50	n=200	n=1000
A	.049 [.046–.052]	.053 [.050–.055]	.052 [.049–.054]
B	.050 [.047–.053]	.050 [.047–.052]	.053 [.050–.055]
C (SVD)	.057 [.054–.060]	.053 [.051–.056]	.055 [.052–.058]
D (SVD)	.060 [.058–.063]	.056 [.054–.059]	.058 [.056–.061]

(For reference, the retracted pinv(X'X) path on the same data: C .417–.422, D .563–.565 at every n — the numerical-implementation failure case of Section 5 Q3, kept out of method comparison.)

On synthetic Gaussian data, all four tests are calibrated (Table 1). On the physics-derived designs the picture splits (Table 2, C/D on the verified SVD path): all four remain within Wilson contact of nominal — A .049–.053, B .050–.053, C .053–.057, D .056–.060, flat in n. The 42–56% breakage this benchmark originally reported for C/D on these tasks was a silent numerical failure of our own harness's pinv(X′X)-based inference path on near-collinear built designs — not a property of the HC3 correction. That failure, its independent verification, and its retraction are reported as a numerical-implementation case in Section 5 Q3; the method comparisons of this paper use the SVD path throughout.

The pooled rates also conceal granularity, which matters for any claim of generality over "physical laws": under the original pinv path the failure was equation-specific, not uniform (5 of 13 tasks at .998–1.00 for C while the other 8 sat at .048–.060); under the SVD path no equation family shows systematic deviation. The Wilson intervals of Table 2 treat all 26,000 fits as independent draws; because seeds repeat within a fixed equation, the honest uncertainty unit for generalization over equations is the task. Task-level bootstrap (resampling the 13 equations; Appendix C) gives the SVD-path C a 95% interval of [.045, .068], against [.197, .707] under the retracted pinv path; leave-one-equation-out moves the SVD pooled rate over [.048, .062], against [.053, .054] for A. One qualification follows. What makes the pinv-failing subset special is design conditioning (Section 5, Q3). An independent-implementation check run in revision (Section 5, Q3) reattributes the failure: a standard SVD-based HC3-Wald implementation (statsmodels) is nominal-compatible on the very tasks where our pinv-based path rejects at 1.00 (8/120 rejections = .067, Wilson [.034,.126] over 60 seeds × 2 tasks — sufficient to reject the old rate of 1.00, not precision evidence of exact 5% calibration), and the two implementations agree to ≥3 significant digits on well-conditioned tasks.

4.2 Power: raw, then calibrated

Raw power at α = 0.05 on S2 H1 (n = 200, pooled over rel ≥ 0.5%): A .52, B .28, C .25, D .68 (C/D SVD). Calibrating every method to a common 5% empirical FPR:

Table 3 — S2 power at matched empirical FPR 5% (C/D on the verified SVD path; thresholds re-derived from SVD p-values on the H0 pool).

Method	n=50	n=200	n=1000
A	.386	.486	.570
B	.187	.254	.389
C (SVD)	.129	.240	.422
D (SVD)	**.494**	**.675**	**.733**

The SVD correction reorders the field. Under the retracted pinv path, C/D "power" was computed against thresholds derived from artifact-contaminated p-value scales (mass piled at 0 and ~1e-195; Section 5 Q3), making those rows meaningless — including the previous draft's claim that A leads at small and medium n. With verified C/D, **D (exog-HC3) has the highest calibrated pooled power at every sample size**, and the cross-equation check confirms it: task-level paired bootstrap (18-task pool, i.30.5 excluded; 10,000 replicates) gives Δ(D−A) = −.080 [−.258, +.081] at n = 50 (crossing zero), **−.170 [−.329, −.041] at n = 200 and −.148 [−.304, −.019] at n = 1000 — intervals that exclude zero in D's favor**; D beats A on 44% of tasks at n = 50 and 72–78% at n = 200/1000. A remains the strongest gate that requires no reference pool (its permutation null is empirically size-valid everywhere measured, §4.7); B is dominated everywhere; C adds nothing over D. The one paired difference outside the four-gate family remains Δ(F−A) = +.177 [+.054,+.323] at n = 50 and +.176 [+.056,+.320] at n = 200 (F wins 63–68% of tasks; F cannot be paired with D without integrating D into the S5 harness — outstanding). τ thresholds describe decision boundaries and are no substitute for these intervals. No method exceeds .73 — most of S2's H1 mass is hard for every test, which Section 4.3 decomposes by family.

Table 4 — S1 blind-spot cartography (raw α, n = 200, rel = 1%).

Scenario	A	B	C	D
cubic (standardized odd term)	.40	1.00	1.00	1.00
interaction ($lx_1 \cdot lx_2$ cross term)	.21	1.00	1.00	.10
opposing curvature (+c, −c, exchangeable)	**.02**	.98	.98	1.00
dilution (one curved variable of two)	1.00	.88	.87	1.00

Three mechanisms, each verified by construction: (i) A's statistic is the mean of squared centered columns — under opposing curvature on an exchangeable design that mean has exactly zero population signal; (ii) a quadratic-projection statistic loses power on odd-symmetric deviations; (iii) B's fitted-value powers are a local Taylor surrogate for any smooth alternative — saturating on interaction where A (.21) and D (.10) struggle — and, as S3 shows, the same surrogate property near-blinds B on rational laws.

4.3 Family-stratified results and sensitivity

Table 5 — S3 calibrated power by family (C/D on the verified SVD path).

Family	A	B	C (SVD)	D (SVD)
exponential	.09	.06	.07	.06
rational	**.65**	.16	.13	**.67**
trigonometric	.66	.69	.58	**.86**

B — near-perfect on synthetic quadratics — is nearly blind to rational laws (.16 calibrated): the fitted-value Taylor surrogate absorbs precisely the low-order curvature by which a saturating law departs from a power law. Under the SVD path D is the strongest gate on both rational (.67) and trigonometric (.86) families; the previous draft's near-zero C/D entries on trigonometric were the pinv artifact (thresholds derived from artifact p-mass), and the S3-level correction agrees in direction with the S2-level one of Table 3. Exponentials defeat every gate (≤.09 — the linearization a log transform induces on them is nearly orthogonal to all four augmentation families); A and D share the rational lead; no method dominates all three families. Every method's strength sits on another method's blind spot; Table 5 has no row-major winner. Sensitivity: scaling every sampling domain ×2 moves exactly one family-method cell by more than 15 points (rational × A: .66 → .84, the expected direction); ×0.5 moves none; heteroskedastic replications agree within .02 on every cell. **These sensitivity readings rest on 30 seeds per cell and cannot be separated from sampling noise at that size**: a single cell at p ≈ .66 carries a 95% interval of roughly ±.17, and the standard error of a difference between two independent 30-seed cells is ≈ .11, so a 15-point trigger cannot distinguish movement from variation. Family × method intervals are now attached (`paper/review3_analysis.py` §3.8b). On the within-cell binomial convention the trigonometric D (SVD) cell is .860 [.848, .871] and rational D is .669 [.655, .683]; on the honest unit — the equation — the same cells read .860 [.592, .998] and .669 [.409, .864]. Rational C moves from .132 [.122, .142] to [.053, .277] the same way. The equation-level intervals overlap across families and are wide enough to contain orderings the point estimates contradict, so the family ranking above is directional, not an ordered effect list; what survives the wider intervals is the sign pattern — D ≥ C on rational and trigonometric, every gate low on exponential — not the point ordering.

4.4 Noise-layer robustness

Table 6 — S1 noise-layer H1 (strict-monomial mean structure) rejection rates at α = 0.05 — false alarms attributable to the noise assumption (n = 200, rel = 1%, 500 seeds; C/D on the verified SVD path).

Scenario	A	B	C (SVD)	D (SVD)
Student-t(3) noise	.034 [.021–.054]	.036 [.023–.056]	.030 [.018–.049]	.030 [.018–.049]
variance growing with prediction	.048 [.032–.070]	.058 [.041–.082]	.050 [.034–.073]	.050 [.034–.073]
AR(1) errors, ρ = 0.6	.038 [.024–.059]	.040 [.026–.061]	.042 [.028–.063]	.042 [.028–.063]
2% point outliers (+8σ)	.040 [.026–.061]	.066 [.047–.091]	.036 [.023–.056]	.036 [.023–.056]
6% group outliers (+5σ)	.038 [.024–.059]	.060 [.042–.084]	.054 [.037–.077]	.054 [.037–.077]

Wilson 95% intervals per cell (500 seeds each). The C/D columns were the last main-text cells still on the pre-correction path; re-computed from the released S1 cells they reproduce the earlier values cell-for-cell (`paper/review3_analysis.py` *§3.8c), which closes that audit item.*

Every method, every non-Gaussian regime: .030–.066, with Wilson intervals in the table (500 seeds per cell). The table's function is exclusionary: the miscalibrations of Table 2 do not come from noisy data; the design, not the noise, is what specification tests are fragile to.

4.5 Do alarms predict extrapolation failure?

Base rate of far-field relative error above 10% across S2 fits: .198.

Table 7 — Alarm precision and recall for far-field error > 10% (S2, all tasks, all n, rel ≥ 0.5%).

Method	precision	recall
A	.39	.63
B	.56	.49

An A alarm raises the probability of severe extrapolation failure from .198 to .39 (2.0×); a B alarm to .56 (2.8×). Of the true-positive catches, 10,560 are A-only and 5,447 B-only against 12,575 shared — the alarm sets overlap barely more than chance. But alarms are ordinal evidence, not magnitude certificates: the suite's two worst extrapolators (i.39.11, i.43.43, both 98% median far-field error) carry the least in-domain curvature and are missed by every method, while ii.38.14 is rejected by A essentially always yet extrapolates benignly (7.7%). Alarm → form evidence; silence → nothing. No rejection is evidence of extrapolation safety.

4.6 The battery: from rejection to routing

Table 8 — Probe: permutation vs wild null by family (n = 200, 200 seeds; entries rel = 1% / 3%).

Truth	perm null	wild null
power (size of power member)	.070 / .075	.075 / .060
rational: false escalation (V = Ø)	**.34 / .31**	.045 / .090
rational: containment	.66 / .70	**.96 / .91**
exponential: containment	.94 / .95	.92 / .95
logarithmic: containment	.87 / .93	.89 / .95
threshold (no battery family): escalation — *correct outcome*	1.00 / 1.00	1.00 / 1.00

With the permutation null, the battery's own rational member rejects its own true family a third of the time — nothing about rational laws is at fault; the null is exchangeability-invalid in (1/x, 1/y) coordinates. The wild bootstrap repairs exactly that and nothing else.

Set-level errors and multiplicity. The battery's decision is the viable *set*, so the errors that matter are set-level, and each member runs at α = .05 without correction — a union decision whose two error types

move oppositely under multiplicity. Two pools, two behaviors. On H0-side cells (truth is a battery member) the reported containment is one minus the exclusion rate of the true family's own member (.0496–.066 across the four truth families; the viable set is never empty *on this pool*), and per-member α does not inflate that rate because exclusion depends on one member only. Inflation appears on the other side: some wrong member survives alongside the truth in 23.3% of power-truth cells (near-neighbour overlap: exponential/logarithmic members retain power data) and 92% of logarithmic-truth cells (rational retains log data) — reported as near-neighbour overlap rather than suppressed in the released tables. On the structural-H1 pool (72,000 cells whose truth is no battery member by design) the viable set is empty — V = Ø, correct escalation — at the rates of Table 10's first column: universally for quadratic/threshold/omitted-variable (1.00), rarely for the interaction/opposing/collinear structures (.001–.05) whose mismatch directions the member families do not span. Routing there means *which wrong family is modal* (Table 10, second column); the battery is a candidate filtering stage whose surviving set the discovery layer must still rank, not a final selection. Downstream selection error after the battery is out of scope here and unmeasured.

Table 9 — Battery on synthetic truths (500 seeds/cell, wild null).

Truth family	Containment (truth ∈ V)	False escalation (V = Ø)
power — size, 8 scenario groups incl. all noise layers	—	.0496 [.046–.053] pooled; per-scenario .043–.054
exponential	.95	.05
logarithmic	.94	.005
rational	.93	.066 pooled (≤ .082 worst rel)

Table 10 — Battery on out-of-family structures (escalation to V = Ø is the correct outcome; column 1 is the V = Ø rate).

Scenario	Correct escalation	Modal false V
quadratic / threshold / omitted variable	1.00 / 1.00 / 1.00	—
dilution	.92	rational
cubic (odd term)	.24	power (.68)
interaction	.05	power (.95)
opposing curvature	**.00**	power+rational (.84)
collinear-quadratic	.03	rational (.61)

Table 10 is Table 4 re-expressed as routing errors: the power member inherits A's documented blind spots, so those structures stay in V under the power label instead of escalating. On the SRSD side the routing is reported descriptively (median |V| 1–2 for 11 of 14 equations; i.30.5 → {exponential} in 78%; ii.38.14 escalates on 67%), because surface-form family labels dissolve under examination — the battery correctly rejects Planck-type "exponentials" that no linearizable family owns.

4.7 The gate comparison: why not just compare against a nested rival?

Table 11 — S5 size on H0 pools (200 seeds/cell; naive and calibrated thresholds for E/F).

H0 pool	A	E naive	E cal.	F naive	F cal.
synth n=50 (600 fits)	.080	.212	.050	.132	.050
synth n=200 (600)	.040	.183	.050	.135	.050
SRSD n=50 (2,600)	.051	.167	.050	.086	.050
SRSD n=200 (2,600)	.038	.152	.050	.123	.050

Table 12 — S5 structural-H1 detection (A at its native threshold; E/F at H0-pool-calibrated thresholds; mean over tasks).

	synth n=50	synth n=200	SRSD n=50	SRSD n=200
A (permutation curvature)	.774	.782	.348	.493
E (CV-quad)	.929	.934	.462	.628
F (AICc-quad)	**.930**	**.940**	**.525**	**.669**

OR(A, E) at each gate's own 5% threshold — a different, larger joint error budget (.065–.102 across the four benches), reported separately: .937/.941/.547/.675. It is not comparable to the rows above and is excluded from this table.

"Matched FPR" applies to three of the four columns: A's own measured size is within Wilson contact of nominal at synth n = 200 (.040), SRSD n = 50 (.051), and SRSD n = 200 (.038), but not at synth n = 50, where A fires at .080 — roughly 1.6× the budget available to E and F. That column is not budget-matched and is reported as measured: F's margin there (.930 vs .774) is read net of A's head start, not as a matched comparison. The same standard the OR row is held to is therefore applied to A: only the three budget-matched columns support a like-for-like ranking.

Three findings. **(a) The obvious alternative is the better default detector, and the margin is exactly A's blind zone.** The quadratic expansion carries per-feature squares *and cross-terms*, so INTERACT (A .24/.26) and OPPOSING (A .03/.02) are caught at 1.00 by both rival gates, and the two SRSD extrapolation-blind equations of §4.5 (i.39.11, i.43.43) are detected at .99–1.00. The direction set, not the null construction, was the binding constraint. CUBIC curvature defeats all three gates equally (~.3): odd curvature is near-orthogonal to every even-column design — the honest general form of the blind-zone lesson.

(b) The rival gates do not come calibrated. Because the rival nests the null, ties and noise break toward escalation: naive thresholds false-alarm at .15–.21 (CV) and .09–.14 (AICc), and calibration to 5% required an H0 reference pool — knowledge a production gate does not have. A's residual-permutation null is an approximation (§3.2) but was empirically size-valid on every pool we measured (.038–.080; synth n = 50 sits at .080 and is reported as measured) without any reference set — A's remaining structural advantage. **(c) Does the calibrated threshold transfer?** Calibrating E and F on all but one H0 equation and evaluating on the held-out equation (13-fold per bench × n) gives held-out sizes of .053/.050 (E, n =

50/200) and .051/.052 (F), with per-equation spread [.010, .095]: reference-pool calibration is not an in-sample artifact — it transfers across equations, though any single held-out equation can land within a ±.04 band of nominal. **(d) The OR union's recall advantage is a joint-budget artifact.** OR(A, E) with each gate at its own 5% runs a joint false-positive budget of .065–.102 across the four benches (Table 11) — its TPR edge over E (+.085/+.047 on SRSD) is purchased with that larger budget, not with information. Recalibrating the *combination* to a 5% joint budget (A's size fixed, E's threshold tightened on the H0 pool) is only defined where A's own size permits it: on synth n = 50, A alone fires at .080 — above the 5% target — so no tightening of E can bring the union to 5%, and the union *is* A there (TPR .774, reported as uncalibratable, not as a 5%-budget cell). Where the recalibration is defined (n = 200, both benches) the union reaches .940/.664 against F's .940/.669 — parity with F, never a gain. reading: at any common error budget F is the frontier; the union is defensible only when a workflow explicitly budgets for its larger naive size. Noise-layer scenarios leave all three gates quiet (.04–.07): the gates measure form, not noise assumptions, and the §4.4 separation survives its most obvious stress test. Cost in fit units: A ≈ 2 full fits + 1 vectorized pseudo-inverse (999 permutations); F = 2 fits; E = 12 — all negligible against the candidate flood a discovery pipeline already fits, which is the actual status quo's cost point: the workflow runs *none* of them today.

4.8 Real data: does the map predict live behavior?

Table 13 — S6 World Bank relations: verdicts (median permutation p over 5 seeds for A; deterministic p for B–D; α = .05).

Relation (target ~ features)	Years all four gates agree	Divergent cells (which gate, which way)
infant mortality ~ GDP pc	4/4 reject	—
CO_2 pc ~ GDP pc	4/4 reject	—
internet penetration ~ GDP pc	4/4 reject ($p < 10^{-16}$)	—
GDP ~ population	4/4 **quiet** (A .71–.91; B .19–.33)	—
life expectancy ~ GDP pc	0/4	A alone rejects 2019 (.035) & 2022 (.008); B alone rejects 2022 (.0002); C/D never
life expectancy ~ GDP pc + health exp pc	2/4 quiet	B rejects 2022 (.0002) while A quiet (.108)

Where signal is strong the gates agree — three relations rejected unanimously every year, one true scaling relation cleared unanimously — and A's five p-values per cell are Monte Carlo replicates of one estimator whose spread is small (median .009, max .094 at resolution 1/1000; no verdict flips beyond boundary cells). Divergence concentrates where the synthetic map puts weak curvature: the life-expectancy cells flip year to year (signal at the detection boundary), and the two-feature cell shows B rejecting where A stays quiet. We label this pattern *consistent with* the dilution mechanism (§4.2, Table 4) rather than a reproduction of it: on observational data the p-value change after adding a second feature can also reflect feature correlation, omitted-variable changes, sample composition, measurement error, cross-country heterogeneity, or spatial dependence — no ground truth exists to separate these, so the claim is signature-consistency, not mechanism confirmation.

Table 14 — S6 dilution probe: the same target with and without a second correlated curved feature (3 years × 5 seeds; detection = median gate verdict).

Relation	A	B	D
cereal yield ~ fertilizer (1 feature)	rejects 3/3 years (.011–.031)	rejects 3/3 (~.003)	rejects 3/3
cereal yield ~ fertilizer + energy pc (2 features)	**quiet 3/3 years** (.067–.149)	rejects 2/3 (.025–.092)	rejects 3/3 (≤.018)

This matches the S1 DILUTION scenario's signature — adding one correlated curved feature silences A on a relation it was rejecting, while the RESET variants continue to alarm — a pattern *consistent with* the dilution mechanism rather than a demonstrated reproduction of it, since feature correlation, sample composition, and omitted-variable changes move the same p-values in the same direction. No ground truth exists on S6 and none is claimed; the contract of §3.6 is signature consistency in both directions: unanimous cells where the map predicts signal, and map-consistent divergence where it predicts weakness.

A companion engineering bench (Appendix E) pushes the same probe across nine physical-engineering sources (147 run cells): unanimous detection of a documented rating-curve misspecification, mechanism-echo divergences matching the synthetic map (small-sample power, interaction carried by a second feature, all three HC3 failure modes live), and mutual misses on locally approximate laws — all recorded, none adjudicated.

4.9 Runtime

Per-dataset wall time (S1 means over 500-seed cells): A 12.0 ms (n = 50) to 25.3 ms (n = 1000); B/C/D 1.0–1.8 ms; the full four-family battery at n = 200 runs in 24 ms; S5 gates E/F 1.5–3.5 ms. All gates are effectively free at the scale of a discovery pipeline that already fits thousands of candidates; the bottleneck resource is not compute but the decision discipline.

4.10 What the full pipeline actually releases

Sections 4.1–4.9 measured the gates in isolation. This section connects the three stages a discovery pipeline actually applies to the *same* fitted model — the fit screen, the specification gates, and the extrapolation consequence — and asks the joint question:

which wrong models does the whole pipeline release, and how dangerous are they? The analysis runs on the S1 quality bench (57,000 fits, 200 seeds) and the S2 pool with recomputed quality (240,000 fits, 500 seeds), C/D on the SVD path throughout (the pinv-based C/D of Tables 2–3 is a numerical-implementation failure case, Section 5 Q3, not a method result). Failed or declined gate evaluations are counted separately and never treated as acceptance. Every number is regenerated by released scripts (`svd_main_results.py`, `no_causal_link.py`) whose anchor checks against previously verified reports fail the build on drift.

Definition (released dangerous model). A fitted model is *released* when it passes the fit screen ($R^2 \geq 0.99$, the threshold a pipeline would actually apply), no gate rejects (A/B native, C/D SVD), and no gate

declines (fail_ *registrations are honest rejections of the evaluation, not acceptance). It is* dangerous* when its far-extrapolation error against the registered ground truth exceeds 10%.

Both thresholds are conventions, and the two stages spend their error budgets on different scales. The 10% danger bound carries no downstream error budget or decision loss — it was fixed before the runs and kept for comparability — so the headline pair (0.67% of screened wrong forms, 6.9% of dangerous ones) is threshold-specific and moves with it. The $R^2 \geq 0.99$ screen is likewise uncalibrated against the gates' 5% error budget: an absolute goodness screen and a significance-level gate are not commensurable, and part of "the gates do not fire at large n" is this asymmetry rather than gate behavior. The sensitivity curve (`paper/review3_analysis.py` §3.3a) runs the threshold over {2, 5, 10, 20, 50}%. At n = 50 the dangerous share of screened wrong forms moves 39.6% → 11.0% → 2.5% across the 2 / 10 / 20% thresholds, and the release rate among screened dangerous models reads 33.4% → 13.2% → 6.4%; at n = 1000 the 20% threshold leaves no dangerous model at all (B = 0). A headline pair therefore moves by roughly a factor of two in each direction over a threshold range a reader would consider reasonable, and by much more at 2%. That is why every rate in this section is printed with its threshold attached and none is offered as a constant. A screen calibrated to the gates' own 5% error budget remains outstanding.

Table 15 — Joint release on the same models (S2: synthetic draws from the registered Feynman equations, not measurements; released as `svd_main_results.md` §C). Every rate names its denominator.

n	Screened wrong forms (A)	Dangerous among them (B)	Released dangerous (C)	C / A	C / B	C / D
50	24,060	2,655 (11.0%)	350	1.5%	13.2%	7.8%
200	23,932	2,296 (9.6%)	106	0.4%	4.6%	0.9%
1000	24,000	2,046 (8.5%)	25	0.1%	1.2%	0.2%

A = wrong forms passing the $R^2 \geq 0.99$ screen; B = those of them whose far-extrapolation error exceeds 10% (dangerous); C = dangerous models that pass the screen and are released by all four gates. D = all dangerous H1 fits at that n, screened or not (12,369 / 12,348 / 12,046). C/B is the release rate among screened dangerous models, C/D among all dangerous models — the two coincide at n = 200 and n = 1000 (106/106, 25/25) but diverge at n = 50, where 616 further dangerous fits that no gate rejected fail the screen and therefore never reach the pipeline's output. Earlier drafts printed C/B in this table and C/D in the running text under the same label, "release rate of dangerous"; both are shown here.

Three readings. **(a) The screen concentrates, the gates miss.** Among wrong forms that pass $R^2 \geq 0.99$, 9.7% are dangerous — the stratum mixing of the previous subsection — and of those dangerous survivors, 6.9% (481/6,997 pooled) walk through all four gates untouched. The joint event is rare *unconditionally* (0.67% of screened wrong forms) but six times more frequent *conditioned on being dangerous*: the wrong forms most worth catching are not the ones most likely to be caught. **(b) The release rate of dangerous forms falls with n — through the screen, not through the gates.** With all dangerous H1 fits as denominator the rate is 7.8% → 0.9% → 0.2% (966/12,369, 106/12,348, 25/12,046); with screened dangerous models as denominator it is 13.2% → 4.6% → 1.2%. Both are reported because they answer

different questions (Table 15). At n = 200 and n = 1000 every released dangerous model has already passed the $R^2 \geq 0.99$ screen (106/106, 25/25): at scale the binding constraint is the fit screen, not any gate. **This gradient is not, by itself, evidence that sample size closes the release channel through detection.** Three effects move the same counts in the same direction — (i) gate power rises with n; (ii) numerator and denominator both change with n; (iii) the best L_2 approximation to a wrong form *within the power-law class* is estimated better as n grows, so its extrapolation error contracts toward a fixed projection error — an approximation-theoretic effect, not a testing property. The dangerous share of screened wrong forms falls 11.0% → 9.6% → 8.5% in the same direction and is consistent with (iii) alone. Decomposing (i)–(iii) requires holding the danger definition fixed across n. Two controls (`paper/review3_analysis.py` §3.2). **(1) Paired label flips.** On the 36,000 (task, rel, seed) triples carrying all three n, the 10% label flips almost symmetrically between n = 50 and n = 200 — 966 triples dangerous only at n = 50 against 945 dangerous only at n = 200 — so at that step the label chases noise around the threshold rather than contracting; the wider steps are one-sided (994 vs 671 from n = 50 to n = 1000; 417 vs 115 from n = 200 to n = 1000).

(2) Fixed danger label. Holding the label at its n = 1000 value and re-stratifying, the release rate among *screened* dangerous models reads 16.8% → 1.9% → 1.2%, and among all dangerous fits under the same fixed label 2.9% → 0.3% → 0.2%. **The gradient survives the control** — it is if anything steeper at n = 50, because the fixed-label denominator drops the fits that merely look dangerous at small n (and those are released least often). So (iii) is real but is not what produces the fall: the fall is the screen and the gates acting on a fixed set of dangerous models, not the danger definition migrating. The defensible reading remains the screen's: a practitioner applying an $R^2 \geq 0.99$ gate at $n \geq 200$ is protected mostly by the screen, and the gates contribute the residual at small n, where the screen is weakest. **(c) The certificate carries information — bounded information.** P(full-green certificate | dangerous wrong form) = .013 against P(certificate | true form) = .845: a dangerous wrong form is 65× less likely to hold the certificate (likelihood ratio .015, exact reading in `svd_main_results.md` §C.3), so the certificate is real evidence — but P(dangerous | certificate) is not zero — 2.2% under the pool's own mixing prior (19 H1 of 32 tasks), 1.5% under a 50/50 prior; the pool-dependent reading is in `svd_main_results.md` §C.3 — and 481 dangerous models hold one. The correct semantics: the certificate shifts the odds substantially and licenses no exclusion.

What the low-fit cases do and do not show. Wrong forms below $R^2 = 0.5$ are jointly missed at 57.5% (n = 50) — but a model at $R^2 = 0.002$ would not survive any sane pipeline's fit screen; its interest is mechanistic (the mismatch direction is orthogonal to every augmentation direction the toolbox spans, ii.11.28), not epidemiological. The joint-release accounting above is the pipeline-relevant quantity; the low-fit silences belong to the gates' blind-spot map (Section 4.4, Table 4).

Direction geometry as diagnostic, one reading among others. Detection correlates with the alignment between the misspecification direction and the test's augmentation space: corr($\hat{y}^2$, residual) against fitted-F detection at Spearman +0.47 (n = 50) to +0.76 (n = 1000), against R^2 at −0.06 to −0.15. We offer this as a *diagnostic*, not an identified mechanism: the alignment statistic is itself built from the same fitted values RESET augments with, so its predictive success is partially circular — an independent test would vary alignment while holding misfit magnitude fixed, and the full augmentation space ($\hat{y}^3$, exog powers, cross-terms), not $\hat{y}^2$ alone, defines the geometry. What the correlation establishes is narrower: detection is predictable from quantities orthogonal to fit magnitude, and no fit statistic dominates that prediction.

The synthetic bench is the control. There, wrong forms passing $R^2 \geq 0.99$ are jointly silent on 0–0.1% of samples (1/811 at n = 50, 0 beyond): synthetic deviations were designed to be detectable, which is why the real-formula release numbers are the finding and the synthetic ones the foil.

4.11 The joint-silence rate: how often does the whole panel stay quiet?

Section 4.10 measured what the pipeline releases *after* the screen. Strip the screen away and what remains is a different quantity: the gate panel's own detection record. That record — not the screen's selectivity — is what a practitioner inherits whenever a fit passes on grounds other than an R^2 threshold: a hand-built correlation, a literature formula being re-certified, a model carried in from an earlier study. The two quantities are not ordered as larger or smaller; they answer different questions, and §4.11 contrasts them only within their own conditioning.

Table 16 — Joint silence on S2's 18 non-monomial H1 equations (2,000 datasets per task per n; 36,000 per n; gates A–D, C/D on the SVD path). Joint silence = no gate rejects AND no gate declines.

n	naive α = .05	H0-pool calibrated
50	37.8%	38.6%
200	29.8%	30.2%
1000	23.2%	23.6%

Three readings.

(a) The synthetic foil does not transfer. On the synthetic bench (S1, same four gates, **no screen applied**) joint silence runs 0.15–0.22% at n = 50. On equations drawn from real physics the same four gates are jointly silent on **37.8%** of wrong forms at the same n — a factor of roughly 170–250 on a like-for-like basis. That like-for-like contrast is the one that carries a claim; comparison against Section 4.10's screened rate would not, and is not made here. The synthetic gallery was built with deviations designed to be detectable; real formulas were not built at all. On this bench this is the benchmark's largest genuine-versus-synthetic gap, and it is offered as such — a gap between two benches, not an estimate of any real-world error rate.

(b) Adding selection gates does not close it. At matched calibration the four-gate panel is silent on **38.6%** of wrong forms at n = 50 (Table 16, calibrated column). Adding the model-comparison gates (CV-quad ∪ AICc-quad on `gate_cmp`'s SRSD arm) raises that to

42.2% rather than lowering it, because the added gates carry their own 5% error budget and the union's silence on true power laws is **90.5%**. The likelihood ratio a silent certificate can carry is therefore only 42.2 / (42.2 + 90.5) ≈ **2.2**×; at a 50/50 class prior the posterior that a silent fit is nevertheless wrong is ≈

32%. Silence is weak evidence, not safety.

(c) The cost of silence is tail-loaded. Among jointly-silent wrong forms at n = 50, far-field error is *small at the median* (0.8%) — which is exactly why the panel is silent — but **7.3%** of silent certificates carry >10% far-field error (p90 = 8.7%). At n = 200 and n = 1000 that share falls to 1.0% and 0.3%. The panel's silence is not uniform in consequence; it concentrates where the misfit is small in-domain and damaging

one decade out.

Relation to Section 4.10 — the denominators must not be pooled. These are not rival estimates of one quantity. Section 4.10's rates are *conditioned on* the $R^2 \geq 0.99$ screen and answer "what does the pipeline emit"; S7's are *unconditioned* and answer "what does the panel alone detect". The n = 50 pair — 13.2% of screened dangerous models released (§4.10, C/B) versus 37.8% of all wrong forms jointly silent (Table 16) — differs in large part because the screen admits only about a tenth of wrong forms (24,060 of 240,000 S2 fits, §4.10) before the gates ever run. Quoting either as "the release rate" without its conditioning would repeat the denominator error Section 4.10 documents, one level up.

5 Analysis

Q1. Does the low-df curvature test beat RESET on univariate quadratic deviations? No — it ties; both saturate from n = 50. What low degrees of freedom buy is calibration, not power — which is what makes A's calibrated numbers trustworthy rather than large. The asymmetries that do appear are structural (odd terms, interactions), where A's quadratic projection is mathematically the wrong instrument.

Q2. Does averaging squared columns miss feature-specific curvature and interactions? Yes, by construction — and the benchmark quantifies when that matters, in three escalating venues: synthetic power cells (Table 4), routing errors (Table 10), and live observational data (Table 14). The dilution mechanism in particular is now a three-level result: constructed (S1), routed (S4), and signature-consistent on observational data (S6). A synthetic blind spot stops being a laboratory artifact the moment it anticipates which real relation goes quiet — though on S6 that correspondence remains pattern-level, not mechanism-proof.

Q3. Does HC3 fix false alarms under heteroskedasticity — and was the observed failure HC3's or ours? Heteroskedasticity was never the disease (Table 6). The observed failure was **ours — a numerical implementation failure, now fully retracted as a method result and reclassified as a numerical-implementation case study**. The method comparisons of this paper (Tables 2, 3, 5, 15) use the SVD path throughout; the pinv(X′X) path survives only here, as the case study. Table 2 (original draft) showed *our* HC3-Wald path breaking on near-collinear built designs; the original draft attributed this to the correction itself ("the robust estimator is the failure mode"). independent-software checks required in revision retract that attribution, and the second round extends them from fitted-HC3 (2 tasks × 60 seeds) to **exog-HC3 across all 13 H0 equations with stratified sampling by equation, sample size, and noise level (176 fits)**, on identical data frames, comparing three paths: our released pinv(X′X) path, our own SVD path (hat and sandwich via pinv(X) of the augmented design), and statsmodels HC3-Wald with the same F reference. Results: (a) the SVD path and statsmodels agree everywhere — maximum $|\Delta p| = .0023$, maximum relative statistic difference .0038, covariance-matrix Frobenius relative difference $\leq 3\times10^{-4}$ — so the mathematical definition is implemented identically and the divergence is numerical; (b) the pinv(X′X) path rejects at rate 1.00 on the 7 tasks whose measured cond(X′X) $\geq 4\times10^{17}$ (measured cond(X) 6.5×10^{8}–3.2×10^{10}), while both SVD paths stay at 0–.10 on every task including those — compatible with nominal, not precision-calibrated; (c) the two quiet tasks with cond(X) ≈ 10 (ii.27.16, ii.27.18) show no failure on any path. Mechanism, stated precisely: the harness calls NumPy 2.5.2 `np.linalg.pinv`, whose default cutoff zeroes singular values below rcond = $10^{-15} \times \sigma_max$(the max(m,n)·ε default belongs to SciPy's pinv, which we do not use); forming X′X squares the condition number, and per NumPy's own

conditioning documentation a condition number 10^k loses ≈ k digits in the worst case — at measured cond(X′X) = 10^{17}–10^{20} the smallest singular values fall below the cutoff and are silently discarded, so the hat diagonal and sandwich are computed from a destroyed spectrum. The earlier draft's informal precision arithmetic is superseded by these measured values. There is no per-call rolling cross-check in the harness — the independent comparison is the released one-off script described above. Corrected conclusion: a silent numerical failure of pinv(X′X)-based inference on squared-condition-number designs; the practical hazard is real (research code commonly pinv-solves; the false alarms are total and silent), but it is an implementation robustness lesson, not evidence against HC3. Accordingly, all method comparisons in this revision (Tables 2, 3, 5, 15) run C/D on the SVD path; the pinv numbers above are the case study, and any residual difference between the paths on suites not yet re-run is an open item (§6).

Q4. Do alarms predict extrapolation error? As binary evidence, usefully (2.0–2.8× enrichment, complementary alarm sets); as magnitudes or guarantees, no. The correct reading is asymmetric: alarm → form evidence; silence → nothing.

Q5. Can the same machinery answer "not a power law — then what?" Yes, as a viability set, after one repair the main benchmark could not have predicted: nulls are coordinate-dependent, and the wild bootstrap is the repair. The formal run delivers the contract (containment .93–.95, pooled size .0496, correct escalation 1.00 on quadratic/threshold/ omitted-variable), with near-neighbour overlap reported rather than suppressed and A's blind spots wearing routing clothes (Table 10).

Q6. Why not just compare against a nested rival? Measured, it wins. F (AICc-quad) leads every cell of Table 12 and removes A's blind spots, because the rival's direction set (per-feature squares and cross-terms) covers the directions A's single mean-square column cannot project onto. The purchase price is a threshold: naive E/F false-alarm at .09–.21 until an H0 reference pool calibrates them; the calibration transfers across equations (leave-one-equation-out held-out sizes .050–.054 on average, §4.7(c)), while A was empirically size-valid on every pool we measured without one (§3.2 records the approximation this rests on). The OR combination, originally reported as the best-recall S5 cell, owes that edge to a larger joint false-positive budget (.065–.102); recalibrated to a 5% *joint* budget it matches but never beats F (§4.7(d)). The portfolio reading follows: a pipeline that can afford a reference pool should default to F; one that must report a p-value per candidate with no reference data should default to A; and a recall-critical deployment should budget for F rather than pay for an uncalibrated union. The benchmark's best test is thereby demoted from default gate to self-calibrating portfolio member, by the benchmark's own protocol; we consider this the paper's most credibility-relevant result, because it is the one a competing paper would have had an incentive to omit.

Q7. Does any of this transfer to real data? Where the synthetic map predicts strong signal, four gates built for physical laws agree on macroeconomic relations they were never designed for (Table 13) — cheap evidence that the map is not a laboratory artifact. Where the map predicts weakness, observational divergence appears with the predicted signature (dilution-consistency, Table 14) — consistent-with, not confirmation-of: too many observational confounds and no truth to adjudicate. What S6 and Appendix E cannot supply — rates, calibration, extrapolation truth, mechanism identification — they do not attempt; real data here is a compatibility probe, not a leaderboard, and their rejection counts are compatibility statistics, not performance rankings.

Q9. Does fit quality certify functional form — or does the full pipeline catch what the screen misses? Neither certificate nor guarantee, and the honest quantity is the joint one (Section 4.10). Fit quality cannot

by itself certify form: within $R^2 \geq 0.99$ the two benches mix true forms (extrapolation error ≈0.1%) and wrong forms (100% >10% error on the synthetic bench — a design property of that bench; 9.7% of screened wrong forms are dangerous on the Feynman-equation pool) with the certificate unable to separate them. The full pipeline releases dangerous wrong forms at 0.67% of screened wrong forms but 6.9% of dangerous ones — the joint miss concentrates exactly on the models most worth catching — and the release rate of dangerous forms falls with sample size (7.8% → 0.2% with all dangerous fits as denominator, 13.2% → 1.2% among screened dangerous models); Section 4.10(b) shows that fall running through the fit screen rather than through any gate. The certificate nonetheless carries real, bounded information: a dangerous wrong form is 65× less likely to hold a full-green certificate than a true one. What no configuration of these tools does is *exclude* the failure mode; certificates of correctness must be anchored in which alternative directions were tested, not in fit magnitude.

6 Limitations

Single candidate family, and no symbolic-regression method (main benchmark). All gates of §3.2–3.5 evaluate a power law; the battery addresses routing, not selection — ranking within V is the discovery layer's job and is not benchmarked. No symbolic-regression method is run anywhere in this benchmark: every gate is applied to a pre-specified candidate, so what is measured is gate behavior on a power-law null, not the pass rate of *method-emitted* wrong forms. The SRBench/SRSD comparison of §1 is therefore an argument for the missing side, not a claim of parity with discovery-side benchmarks, and a suite that attaches these gates to at least one SR method's output is the next step — named here so that no reader mistakes this paper for having taken it.

One rival, one criterion each (S5). The rival is the quadratic expansion and the criteria are 5-fold CV and AICc — the reviewer's canonical objection, not a survey of selection gates; LOOCV, held-out likelihood, or non-nested rivals may rank differently, and the calibration asymmetry (reference pool vs self-calibrating null) is a workflow property whose cost we price but do not optimize. **Battery scope.** n = 200 only; no member for interaction or trigonometric structures; SRSD side descriptive pending a symbolically exact linearizability registry. **Real-data scope (S6).** Six relations, one source, four years, no ground truth: no rate claims are possible, the relation list is preregistered to prevent cherry-picking but remains six choices, and the year-to-year verdict flips in weak-curvature cells signal genuinely boundary-level signal rather than gate defect — these cells are reported as measured, not smoothed. **Fixed augmentation.** Built powers fixed at (2, 3). **Log-space scope.** All gates operate in log-log space; additive-error models are out of scope and marked not-applicable. **Statistical precision and clustering.** S1/S2/S4 rates rest on 500 seeds; S3 on 30 (directional); S5 on 200; S6's 5 seeds are Monte Carlo replicates of one estimator, not samples (median p spread .009, max .094). Seeds repeat within equations, so Table 2's Wilson intervals understate cross-equation uncertainty; task-level bootstrap intervals and leave-one-equation-out ranges (Appendix C) are the honest units for generalization over equations, and the HC3 failure is concentrated in 5 of 13 equations. No multiple-comparison correction is applied across the 1,480 cells; rates are descriptive, and family-level set decisions are analyzed as sets (§4.6) rather than as four independent tests.

Joint-silence suite (S7). The suite's two arms rest on different benches — A–D on the S2 18-equation pool (36,000 datasets per n) and the model-comparison extension on `gate_cmp` (3,800) — so their levels are reported separately and are **not** comparable across arms; only within-arm contrasts are used. The

four-gate arm's H0 reference was not re-measured on the S7 seed grid, so Table 16 is an H1-side measurement: the 90.5% true-form silence rate quoted in §4.11(b) belongs to the **model-comparison arm alone** and is not the four-gate arm's null. Table 16 carries no Wilson intervals — the 2,000 datasets per task are draws within 18 equations, so equations, not datasets, are the independent unit, and the per-task spread is wide by design (per-task detail is retained rather than macro-averaged away). Two exclusions predate the suite: `feynman-i.30.5` (withdrawn, v5 App. B) and `rel = 0` cells (no certificates exist there). **Provenance.** S7's cached outputs are `runs/joint_silence/`. The summary table in that directory's `REPORT.md` disagrees with the per-task CSV it was computed from (32.0% vs 37.8% at n = 50 after weighting; 70% vs 90.5% for the H0 baseline) and carries a dated in-file annotation to that effect. Every number in Table 16 is regenerated from the per-task CSV (`joint_silence_srsd_per_task.csv`), and the discrepant summary is not used anywhere in this paper.

HC3 independent verification. After two revision rounds: fitted-HC3 (C) is verified on 2 failing tasks × 60 seeds; exog-HC3 (D) is verified across all 13 H0 equations with stratified sampling (176 fits) — in both cases our SVD path matches statsmodels to $\leq .0023$ in p and $\leq 3\times10^{-4}$ in covariance, while the X′X path fails only at measured cond(X′X) $\geq 4\times10^{17}$. This revision goes further than the previous one: all main-text method comparisons (Tables 2, 3, 5, 15) run C/D on the SVD path, and the previous draft's path-specific footnotes are replaced by re-derived numbers. The S1 noise-layer rows of Table 6 — the last main-text C/D cells still on the pre-correction path — are now re-run on the SVD path and reproduce the reported values cell-for-cell (§4.4), with Wilson intervals attached. Still outstanding: the S5 integration of D, and regeneration of the Appendix D figure panels (below). **Permutation validity.** The residual-permutation null is an approximation whose finite-sample behavior under leverage, dependence, or stronger heteroskedasticity is unmeasured here; leverage-adjusted, wild, block, cluster, and fixed-design null variants are the required stress set. **OR budget.** Union gates are reported at both naive and joint-5% calibration; any deployment comparison must fix the budget first (§4.7(d)). **Threshold conventions and the n-gradient (§4.10).** The 10% danger bound and the $R^2 \geq 0.99$ screen are conventions fixed before the runs, not calibrations against a downstream error budget, and the two stages therefore spend their error budgets on incommensurable scales. Both headline joint-release quantities are threshold-specific. The release gradient across n mixes at least three effects — rising gate power, a changing numerator and denominator, and the contracting best L_2 approximation to a wrong form within the power-law class — and this paper does not separate them: the threshold-sensitivity curve and the n-decomposition are registered as outstanding (`paper/review3_analysis.py`), and §4.10's gradient claim is weakened to the screen's binding-constraint reading for that reason. **Figure provenance (Appendix D).** The Appendix D panels are regenerated on the verified SVD path (`paper/review3_analysis.py`): D.1 plots the H0 p-value CDFs of both pools, D.2 the per-task calibrated power, D.3 calibrated power versus n with task-level intervals, and D.4 the alarm precision/recall and error distribution of §4.5 — D.4's base rate (0.198) and A/B precision-recall reproduce Table 7 exactly. The pre-correction figure set, whose captions had drifted from the published images, is retired; D.5 (timing) is unchanged because timing does not depend on the inference path.

7 Reproducibility statement

Every generator is seeded from a documented rule embedding the grid position, so any *numeric* cell of any table regenerates bit-identically, and the Appendix D panels are regenerated from the released cells by `paper/review3_analysis.py` (Section 6, "Figure provenance"). The harness records content hashes of the code under test — including the exact `form_check` implementation — and of the harness itself; a statistic-mismatch sentinel verifies on every call that the wrapper's recomputation agrees with the native implementation. The battery is implemented at the experiment layer over the unmodified core code path, so audited core-code hashes remain valid. S5's gate definitions and grid are frozen in `gate_compare.py`'s preregistration docstring (seed bases disjoint from all prior suites). S6 pulls from the World Bank API with every raw response cached under the run directory (`runs/wb_real/raw/`), making the observational suite reproducible against the cached snapshot and refreshable against the live API; the relation and probe lists are frozen in `wb_run.py` / `wb_blindspot.py` before any gate ran. The revision's re-analyses are scripted and released with their raw inputs: `paper/revision_analysis.py` regenerates every number in §4.1's task-level paragraph, §4.6's set-level errors, §4.7's OR-budget and transfer analyses, and §4.8's Monte-Carlo spreads from the released per-cell parquet/CSV files; `paper/verify_round2.py` regenerates the second-round D-verification, Wilson and paired-bootstrap numbers; the independent HC3 check (Q3) uses statsmodels 0.14 on identical data frames. The joint-release analysis of Section 4.10 and the SVD-path Tables 2/3/5 numbers are regenerated by `svd_main_results.py` (from runs/cd_svd and the released quality benches, anchor-checked against the verified summaries). The no-entailment analysis of Section 4.10 (and its twelve anchor checks against the previously verified reports) is regenerated by `no_causal_link.py` from the released quality benches (`runs/quality50/`, `runs/srsd_full/quality_recompute.parquet`) and the SVD-path C/D rerun (`runs/cd_svd/`), writing `no_causal_link.md` and `figures/no_causal_link.png`; the anchor list includes the joint-silence rates of the companion audit, so silent drift between the paper's two evidence chains fails the build. The engineering companion bench of Appendix E is released the same way (`real_data/`): raw snapshots cached with SHA-256 hashes and access dates (2026-09-06), a manifest of preregistered relations with per-relation notes and per-source license fields, and replayable fetch/prepare/smoke scripts; every declined evaluation carries a typed reason. The benchmark artifact (repository archive with all generators, per-cell results, and content-hashed configurations) is deposited with the publication; a permanent DOI is assigned at release. All cells' raw results (parquet and CSV), failure registrations with typed reasons, selection and exclusion tables, analysis scripts, and figure sources are released. Zero-noise cells deliberately exercise the failure-registration path; the battery's 94,000 datasets and S5's 20,400 datasets contain zero data-layer failures, which is itself a registered outcome.

8 Conclusion

We asked what the fit-quality metrics that certify models in discovery pipelines can and cannot deliver, and measured both directions on two benches (508,825 fitted datasets plus a quality re-evaluation over 297,000 of these fits). Fit quality cannot by itself certify a functional form: within the single $R^2 \geq 0.99$ stratum, true forms and wrong forms share the certificate while their extrapolation behavior differs by two orders of magnitude. The evaluated specification gates share blind spots: on real formulas, dangerous wrong forms (extrapolation error >10%) that pass the $R^2 \geq 0.99$ screen are jointly missed by all four gates at 6.9% — the joint miss concentrates on the models most worth catching — and the release rate of

dangerous forms falls with sample size (7.8% → 0.2% with all dangerous fits as denominator, 13.2% → 1.2% among screened dangerous models), a fall Section 4.10 attributes to the fit screen rather than to any gate. The certificate carries real, bounded information (a dangerous wrong form is 65× less likely to hold one than a true form); what it cannot do is exclude the failure mode. These are bounded claims: we quantify the extent, frequency, and sample-size scaling of the blind spots — not the absence of a statistical relationship, which our own likelihood-ratio accounting contradicts.

The falsification-side benchmark built to measure this is itself a deliverable: seven suites, one protocol that puts every gate on a common basis, and full registration of every failure — with all C/D method comparisons on the independently verified SVD implementation. Under that correction the gate ranking changes substantively: D (exog-HC3, SVD) has the highest calibrated pooled power at every sample size on the Feynman-equation pool, with the cross-equation bootstrap confirming the advantage at $n \geq 200$ — the previous draft's small-sample lead for A was an artifact of retracted numerics. The benchmark's most striking claim did not survive its own revision: the 42–56% HC3 false alarms were a silent numerical failure of a pinv-based inference path on near-collinear built designs — an SVD-based implementation is calibrated exactly where it fails — a retraction that is itself the benchmark working as designed, and the reason this revision replaces rather than footnotes the affected numbers. Raw power rankings still invert across families: no gate covers all three families of Table 5 — exponentials defeat every gate — and the nested quadratic rival adjudicated by AICc remains the strongest default where a reference pool exists — paying for it with a naive 9–14% false-alarm rate until a reference pool calibrates it — while the permutation curvature gate remains the strongest gate requiring no reference pool, and the OR union's apparent recall premium is a joint-error budget artifact that vanishes at a common budget. The battery extends the machinery from rejection to routing, after a one-line null repair that carries a general lesson: a null distribution's validity is a coordinate-dependent property. And on macroeconomic data the map does what a map is for: it predicts where four gates will agree, and which gate goes quiet on which relation.

For the pipeline designer the results compose into one rule: certificates must be anchored in which alternative directions were tested and found silent — never in fit magnitude, and never in the absence of alarms from a gate set whose blind directions have not been mapped. In scientific discovery, the model you release on the strength of its fit and its clean diagnostics is the one that breaks your extrapolation.

References

[1] Ramsey, J. B. (1969). Tests for specification errors in classical linear least-squares regression analysis. *Journal of the Royal Statistical Society, Series B*, 31(2), 539–551.

[2] White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4), 817–838.

[3] MacKinnon, J. G., & White, H. (1985). Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics*, 29(3), 305–325.

[4] Long, J. S., & Ervin, L. H. (2000). Using heteroskedasticity consistent standard errors in the linear regression model. *The American Statistician*, 54(3), 217–224.

[5] La Cava, W., Orzechowski, P., Burlacu, B., de França, F. O., Virgolin, M., Jin, Y., Kommenda, M., & Moore, J. H. (2021). Contemporary symbolic regression methods and their relative performance.

NeurIPS Datasets and Benchmarks Track.

[6] Mundhenk, T. N., Landajuela, M., Glatt, R., Santiago, C. P., faissol, D. M., et al. (2022). Rethinking symbolic regression datasets and benchmarks for scientific discovery. *arXiv:2206.10540*.

[7] Udrescu, S.-M., & Tegmark, M. (2020). AI Feynman: A physics-inspired method for symbolic regression. *Science Advances*, 6(16), eaay2631.

[8] Bakarji, J., Champion, K., Kutz, J. N., & Brunton, S. L. (2022). Discovering governing equations from partial measurements with deep delay autoencoders. *Nature Computational Science*.

[9] Cranmer, M. (2023). Interpretable machine learning for science with PySR and SymbolicRegression.jl. *SciPy Conference*.

[10] Liu, R. Y. (1988). Bootstrap procedures under some non-i.i.d. models. *Annals of Statistics*, 16(4), 1696–1708.

[11] Davidson, A. C., & Flachaire, E. (2008). The wild bootstrap, tamed at last. *Journal of Econometrics*, 146(1), 162–169.

[12] Burnham, K. P., & Anderson, D. R. (2002). *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach* (2nd ed.). Springer.

[13] Arlot, S., & Celisse, A. (2010). A survey of cross-validation procedures for model selection. *Statistics Surveys*, 4, 40–79.

[14] World Bank (2026). World Bank Open Data API, indicators NY.GDP.MKTP.PP.CD, NY.GDP.PCAP.PP.CD, SP.POP.TOTL, SP.DYN.LE00.IN, SP.DYN.IMRT.IN, EN.GHG.CO2.MT.CE.AR5, EN.GHG.ALL.MT.CE.AR5, SH.XPD.CHEX.PC.CD, IT.NET.USER.ZS, AG.YLD.CREL.KG, AG.CON.FERT.ZS, EG.USE.PCAP.KG.OE. https://api.worldbank.org (cached snapshot released with the benchmark).

[15] UCI Machine Learning Repository. archive.ics.uci.edu (datasets: auto_mpg, automobile, concrete, slump, energy_efficiency, yacht, ccpp, air_quality, gas_turbine, tetouan, steel). Accessed 2026-09-06; raw snapshots and SHA-256 hashes released with the benchmark; redistribution per each dataset's stated license.

[16] NIST Statistical Reference Datasets (StRD). itl.nist.gov/div898/strd. Accessed 2026-09-06; public domain.

[17] NIST Chemistry WebBook, SRD 69. webbook.nist.gov. Accessed 2026-09-06; public domain (U.S. Government work).

[18] U.S. Geological Survey. National Water Information System (NWIS instantaneous values, 12 gauges, WY2023) and FDSN event service (global M ≥ 5.5, 2000–2024). earthquake.usgs.gov / waterdata.usgs.gov. Accessed 2026-09-06; public domain (U.S. Government work).

[19] Open-Meteo historical weather API (ERA5 reanalysis), 10 wind sites, 2023. open-meteo.com, CC BY 4.0 (ERA5: Copernicus licence). Accessed 2026-09-06; reanalysis, not in-situ measurement.

[20] OpenML. openml.org (datasets: ailerons, cpu_act, elusage). Accessed 2026-09-06; per-dataset licenses vary and are recorded in the released manifest.

[21] U.S. Department of Energy, fueleconomy.gov vehicle fuel-economy data, model years 1984–2025. Accessed 2026-09-06; public domain (U.S. Government work).

[22] RDatasets (Vincent Arel-Bundock), datasets: pressure, cars, rock, trees, attenu, housing, electricity, stormer, whiteside, cement, petrol, women, airquality, puromycin. vincentarelbundock.github.io/Rdatasets. Accessed 2026-09-06; licenses inherit from each source package (mostly GPL/BSD) and are recorded in the manifest.

Appendix A — Suite composition

Suite	Cells	Fits per cell	Total fits	Extra evaluations
S1 synthetic	285	500	142,500	—
S2 SRSD-Feynman	480	500	240,000	240,000 test sets; 232,500 extrapolation probes
S3 family-stratified	392	30	11,760	—
S4 battery	188	500	94,000	—
S5 gate comparison (v3)	102	200	20,400	—
S6 real data + probe (v3)	33	5	165	—
Total	**1,480**	—	**508,825**	**472,500**

Outside the seeded totals: the Appendix E engineering bench contributes 150 single-realization smoke cells (147 run), one fit per gate per cell.

Appendix B — Failure registration taxonomy

Reason	Meaning	Exercised by
`degenerate_zero_residual`	Zero-noise cell on exact ground truth; test undefined	S1/S2 $\sigma = 0$ cells
`rank_deficient`	Design matrix numerically rank-deficient	collinear designs
`insufficient_df`	$n - k < 1$ after augmentation	$n = 50$ multi-feature cells
`hc3_leverage_degenerate` / `hc3_singular_covariance`	HC3 undefined	degenerate built columns
`statistic_mismatch`	Wrapper statistic ≠ native statistic (drift sentinel)	continuous audit
`checked_false_*`	Native test declines to run (non-positive data, $n < 12$, non-finite)	not-applicable tasks
`runner_error` / generation failure	S5 SRSD-side generation exceptions	S5
indicator archived / no data	S6 indicator not published for a year	S6 (`EN.ATM.CO2E.KT` → AR5 replacement, logged)

All registered failures remain in the audit trail; denominators are documented per table. The battery's 94,000 and S5's 20,400 datasets contain zero data-layer failures.

Appendix C — Headline Wilson 95% intervals

S2 H0, n = 200 (26,000 fits per cell; C/D on the verified SVD path): A .053 [.050–.055]; B .050 [.047–.052]; C .053 [.051–.056]; D .056 [.054–.059]. S1 H0, n = 200 (4,500 fits per cell): A .037 [.032–.043]; B .041 [.036–.047]; C .045 [.039–.051]; D .042 [.036–.048]. Battery power-family size: .0496 [.046–.053]. S5 H0 pools (600–2,600 fits per cell): intervals within ±.013 of every Table 11 entry. Intervals for all other cells are computable from released per-cell counts.

Task-level uncertainty (S2, n = 200). Seeds repeat within an equation, so fit-count intervals understate cross-equation uncertainty; the equation is the honest unit for generalization over physical laws. Resampling the 13 H0 equations against the SVD-path p-values (10,000 bootstrap replicates; `paper/review3_analysis.py` §3.8a): C .053 [.050, .056] and D .056 [.055, .058], against A .0535 [.051–.056] and B .0497 [.048–.052] as reported above; leave-one-equation-out moves C over [.052, .054] and D over [.056, .057]. All four gates' cross-equation intervals are of the same order on the corrected path — the wide interval belongs to the pinv path (below), not to the test.

The same two statistics under the retracted pinv path read C .417 [.197–.707] and D .563 [.276–.782], with leave-one-out ranges [.368, .448] and [.527, .606]. This is the appendix's widest swing and the clearest single demonstration that a task-level interval can be a property of the numerical path rather than of the test: the path-conditional width for C (.51) exceeds the SVD path's cross-equation width (.02) by more than an order of magnitude. Under that path C's per-equation rates split 5 equations at .998–1.00 (i.34.27, i.38.12, ii.27.18, iii.12.43, iii.15.14) against 8 at .048–.060, and no single equation contributes more than 18.5% of its false alarms — the failure was widespread across the near-collinear subset, not one bad task. All method comparisons in this paper run on the SVD path (Section 5, Q3).

Macro and micro averages coincide here only because the design is balanced (2,000 fits per equation); on unbalanced pools they should be reported separately.

Appendix D — Result figures

Machine-generated figures, one per figure claim of the main text, regenerated on the verified SVD path of this revision (`paper/review3_analysis.py`; image sources under `figures/`). Each caption states the suite, the sample, and the inference path the panel is drawn from.

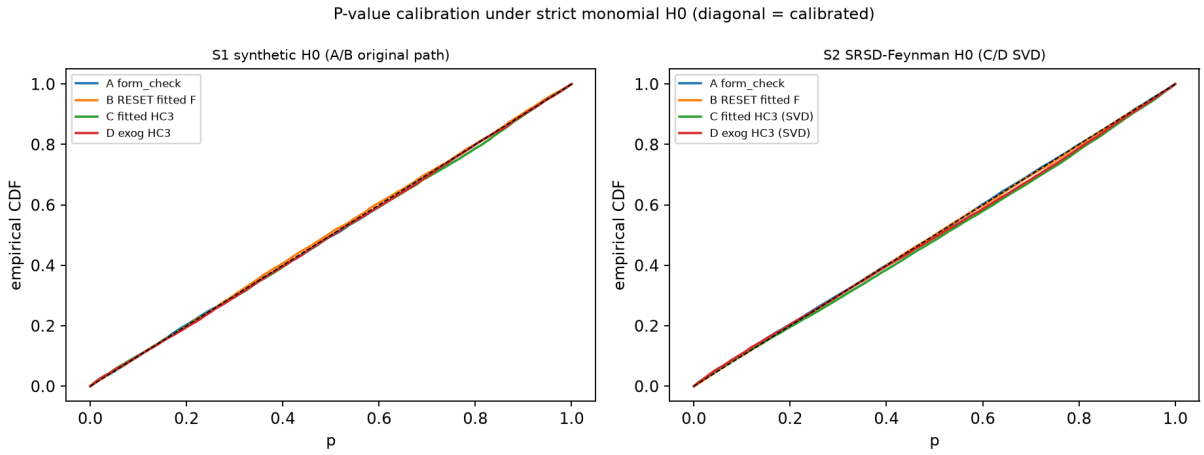


Figure D.1 — Empirical CDF of p-values under strict monomial H0 (uniform = calibrated): S1 synthetic H0 (left) and S2 SRSD-Feynman H0 (right). All four gates track the diagonal on both pools — the calibration claim of Tables 1–2 in graphical form.

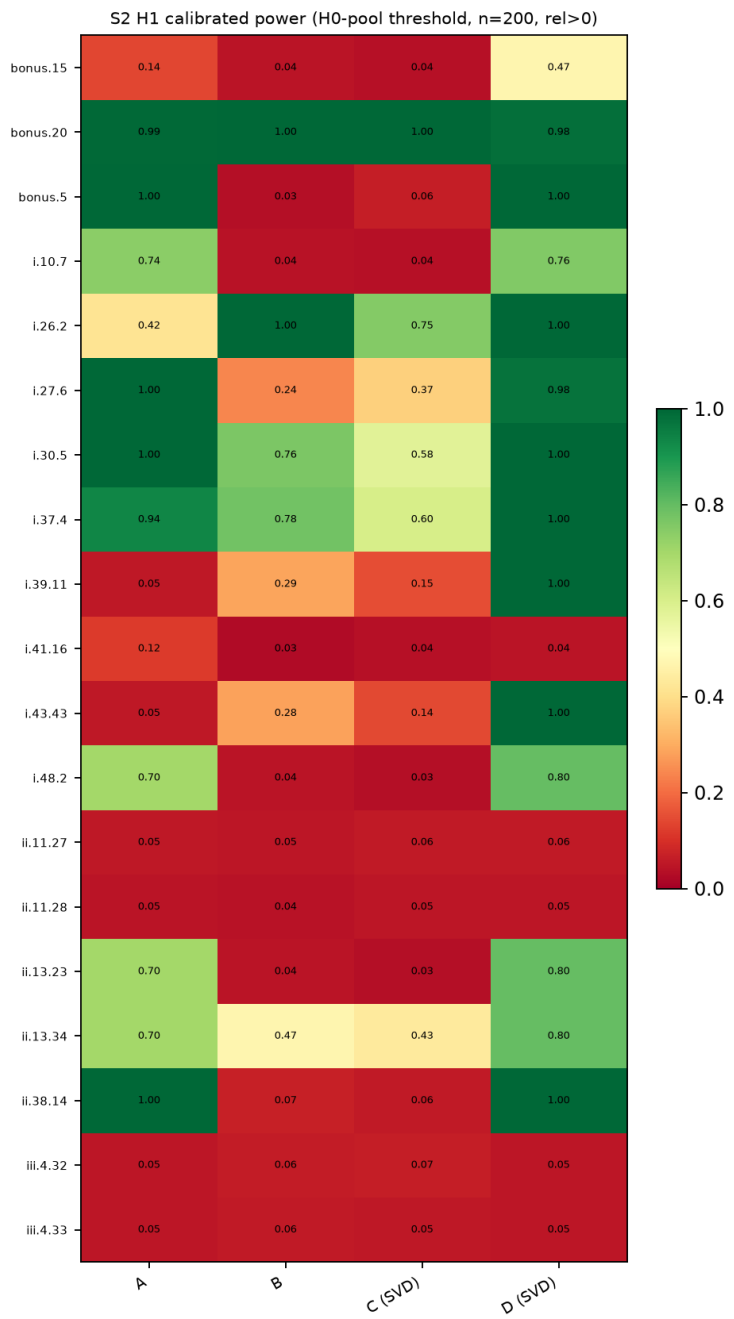


Figure D.2 — S2 H1 calibrated power per task (H0-pool 5% threshold, n = 200, rel > 0): one row per task, one column per gate. The two extrapolation-blind equations of §4.5 (i.39.11, i.43.43) are dark under A and B but reach 1.00 under D; iii.4.32 / iii.4.33 stay dark for every gate.

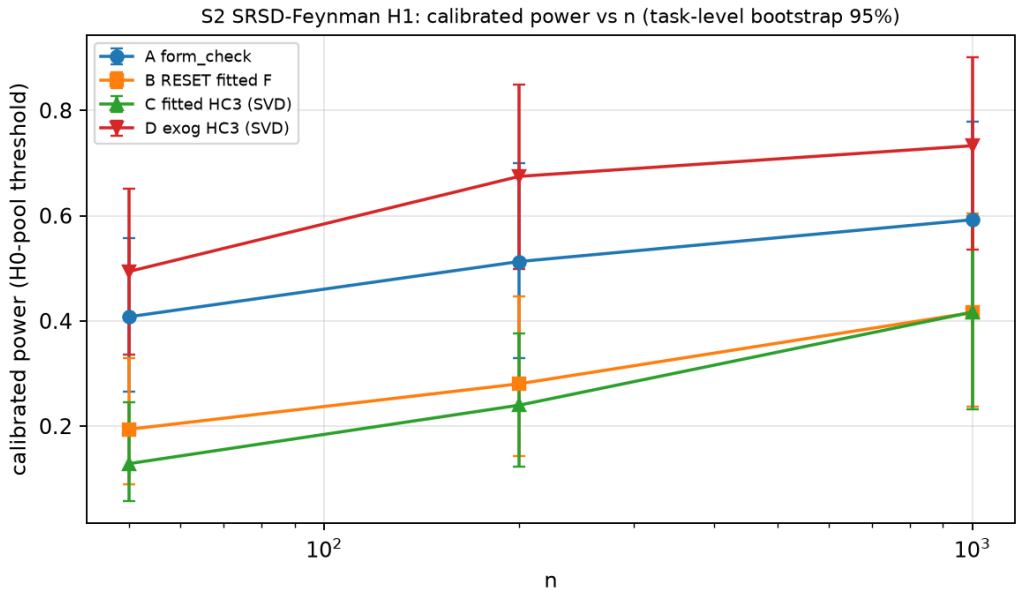


Figure D.3 — S2 H1 calibrated power versus n, with task-level bootstrap 95% intervals (10,000 replicates over the task pool). D leads at every n; the B and C intervals overlap.

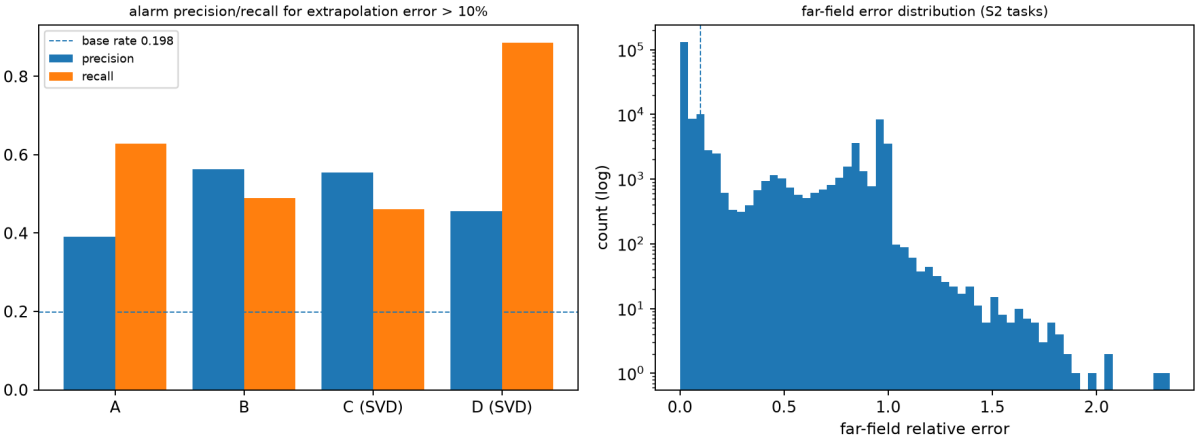


Figure D.4 — Left: alarm precision and recall for far-field error > 10% against the 0.198 base rate (S2, all tasks, all n, rel > 0); A trades precision for recall, D the reverse. Right: the far-field error distribution those rates are computed on.

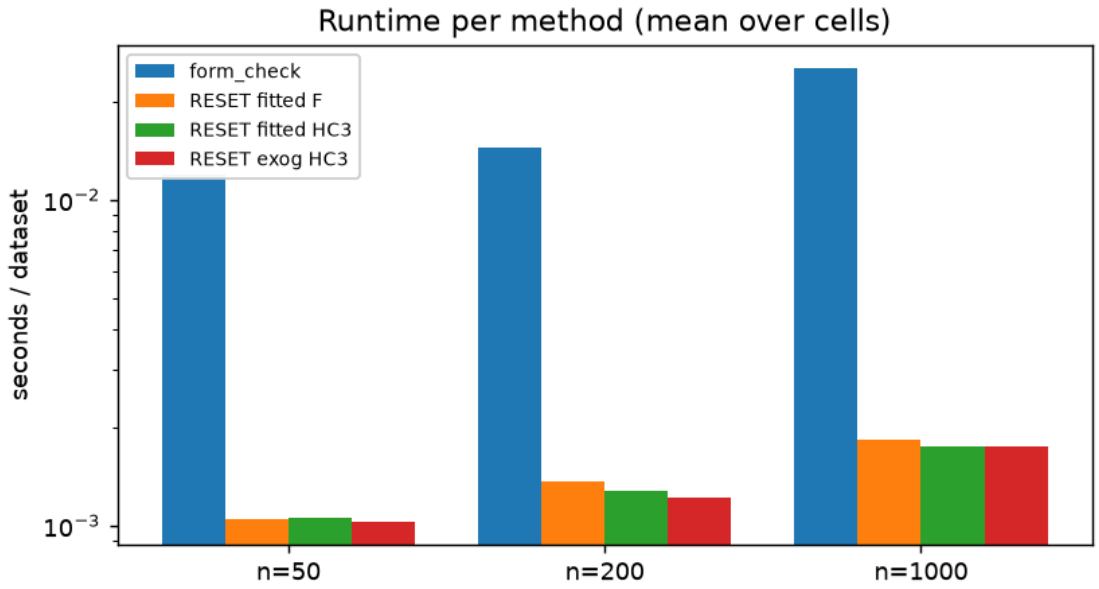


Figure D.5 — Per-dataset wall time; all four tests stay under 30 ms across n; timing does not depend on the inference path.

Appendix E — The real-engineering data bench (companion to S6)

S6 probes mechanism transfer on macroeconomic aggregates. The deployment domain these gates are actually built for is physical-engineering measurement — fluid, civil, hydraulic, energy, meteorological, seismic, chemical, and aerospace data, where power laws are the daily candidate models. This appendix documents a companion bench of that kind, run under the same contract as S6: real data has no ground truth, so the bench registers compatibility and divergence, not error rates; its cells are single realizations and stand outside Appendix A's seeded totals.

Composition and discipline. Nine open, license-free sources; 91 datasets; 150 candidate log-log relations preregistered in the released manifest with per-relation physical notes; 761,515 raw rows (Table E1). Raw bytes are cached (~115 MB) with SHA-256 hashes; cleaning is replayable and imputation-free — rows with missing or non-positive values are dropped and logged, and the few unit conversions are explicit in the manifest (e.g., Whiteside °C → K). Simulator datasets that violate the real-data discipline (a marine-engine and two wave-energy numerical models) are excluded and listed, as are WebBook fluid pages below the 10-row rejection-detection floor; one gauge whose instantaneous series was never published is logged as an empty skip. ERA5 cells are marked as reanalysis, not direct measurement.

Table E1 — Composition by source.

Source	Datasets	Relations	Rows	Examples
USGS NWIS rating curves	12	12	442,269	discharge ~ gage height, 12 gauges
UCI	11	38	145,920	CCPP, Tetouan power, steel, yacht, concrete
EPA fueleconomy	1	4	48,624	CO_2 ~ MPG; combined MPG ~ displacement
Open-Meteo / ERA5	10	20	87,600	100 m ~ 10 m wind speed (shear); gust factor
OpenML	3	8	21,997	aileron position; CPU utilization
USGS FDSN	1	3	12,334	felt reports, stations, intensity ~ magnitude
RDatasets	14	26	1,349	cars, trees, puromycin, stormer, rock
NIST StRD	13	13	968	certified nonlinear calibration curves
NIST WebBook	26	26	454	phase-transition enthalpies ~ temperature
Total	**91**	**150**	**761,515**	—

Protocol. Each cell applies the §3.2 gates unchanged at $\alpha = 0.05$ — A with its $B = 999$ permutation p, B/C/D as registered — on the positivity-filtered relation, subsampled to $n \leq 800$ (seed 42) to bound permutation cost. The run is a smoke pass: one realization per cell, compatibility plus first verdicts, not a rate estimate. Of 150 cells, 147 run; three are registered `insufficient_n` by design (Bennett5 and Roszman1 have no positive rows after the log-space positivity filter; Danwood has $n = 6$) — the bench's honest-failure probes working as intended.

Table E2 — Smoke verdicts on the 147 run cells.

Gate	Rejections	Registered failures
A — permutation curvature	97/147 (.660)	10 declines, all `checked_false_small_n` ($n \leq 11$)
B — RESET fitted F	111/147 (.755)	0
C — RESET fitted HC3	108/147 (.735)	0
D — RESET exog HC3	94/147 (.639)	3: `insufficient_df` ($n = 13$, four features), `hc3_leverage_degenerate` ($n = 546$), `rank_deficient`

A and B agree on 123/147 cells (92 both reject, 31 both quiet; 5 A-only, 19 B-only). The failure columns are themselves informative: D's three declines are exactly the three HC3 failure modes of Table 2 and Appendix B, now on observational data; A's ten are uniform declines-to-run on $n \leq 11$ cells; B and C never fail. Nothing fails silently — every declined evaluation is registered with a typed reason.

Designed regimes. The bench's design document assigns expected regimes to 66 of the 150 cells before the smoke pass (Table E3). The assignment is a reading aid, not a scored test: the H0-side laws are approximate by design, and at n up to 800 an approximate law is a detectable null. The 12 rating-curve cells — whose registered form $Q \sim h$ omits the documented stage offset h_0 of the true rating law $Q = C(h - h_0)^n$, a mismatch noted in the manifest — are rejected unanimously by A and B alike (12/12): detection of a documented misspecification, not a false alarm. The five H0-side cells A retains are those with no designed departure mechanism — the braking-distance v^2 law, the tree allometry, the road-load relation

— plus the two shear sites with the smallest fitted curvature.

Table E3 — Designed-regime cells, rejections at α = 0.05.

Designed regime	Cells	A	B	C	D
Approximate power laws (H0-side)	30	25	27	23	23
Known non-power-law physics (H1-side)	36	20	24	24	14

(A's H1-side count includes its nine WebBook declines at n ≤ 11 as non-rejections.) On the H1 side no gate sweeps the board: the misses concentrate in WebBook enthalpy cells whose measured temperature window keeps the local power-law approximation good — e.g., toluene Δ_vapH ~ T: all four gates quiet at n = 26 — a mutual-miss cell recorded as measured.

Table E4 — Named cells: mechanism echoes and divergences; p-values at α = 0.05 in A / B / C / D order, — = declined with registered reason.

Cell	A / B / C / D	Reading
Wind shear v_{100} ~ v_{10}, 10 ERA5 sites (n = 800 ea.)	8/10 / 10/10 / 7/10 / 7/10	B flags every site ($p \le 2\times10^{-4}$); A retains exactly the two smallest-curvature sites (.23, .41) — a graded response, not a threshold artifact
Gust factor ~ v_{10}, same sites (n = 800 ea.)	10/10 ×4	unanimous rejection at every site; no truth to adjudicate
puromycin rate ~ conc (Michaelis–Menten), n = 23	**.011** / .069 / .169 / .169	A detects the saturation curvature, B misses at α = .05 — Table 3's small-sample power gap, live
stormer Time ~ Viscosity + Wt (calibration law), n = 23	.001 / 7×10^{-4} / 9×10^{-3} / —	single-feature cells quiet (A .94, B .99); the second feature carries the interaction the law contains
EPA CO_2 ~ MPG (stoichiometric inverse, exponent −1), n = 800	.030 / 1.5×10^{-4} / .727 / .727	A and B reject on a near-exact physical law bent by fleet drive-mix curvature
EPA combined MPG ~ displacement (road load), n = 800	.386 / .596 / .599 / .599	designed power law passes
CCPP PE ~ V, n = 800	.642 / 8×10^{-4} / ≈0 / .023	proxy curvature: V correlates .81 with the strongly curved AT, so B/C's fitted-power columns inherit AT's curvature while A's V-only direction set cannot — the mirror of dilution
automobile horsepower ~ engine size, n = 203	.878 / 1.2×10^{-4} / .094 / .094	divergence under heteroskedastic body-style scatter
rock perm ~ area + peri, n = 48	.835 / 1×10^{-5} / 5.6×10^{-6} / .596	A/D quiet, B/C reject; no truth to adjudicate
housing price ~ 4 features, n = 546	.723 / .683 / ≈0 / —	D's leverage-degeneracy failure mode, live at n = 546, while A/B/C return verdicts

Scope. What the bench supports is what S6 supports: compatibility at scale (147/150 cells run across nine sources, zero silent failures), failure registration on observational data, and mechanism-echo cells whose directions match the synthetic map. Its per-gate rejection totals (97/111/108/94 of 147) are compatibility statistics on single realizations without ground truth — they rank nothing, and reading them as performance would repeat exactly the leaderboard confusion the benchmark exists to correct — A's calibrated conservatism (the shear sites, the road-load law, puromycin at n = 23), A's interaction and dilution behavior (stormer, CCPP), D's structural fragility (all three failure modes live). What it cannot support: rates, calibration, or rankings — single realizations, no ground truth, and time-ordered NWIS/Open-Meteo series whose permutation-independence assumption is an approximation marked in the manifest. All four p-values per cell are released (`real_data/smoke_results.csv`) with the manifest (`real_data/clean/manifest.json`, SHA-256 per source) and replayable fetch/prepare/smoke scripts.

Appendix F — Core code listings

Abridged reference implementations (error handling and instrumentation omitted; full source released with the benchmark). Listing 1 is the wrapper around the project's native `form_check`: it recomputes the curvature statistic in log-log space, verifies it against the native value (drift sentinel), and derives the empirical p-value from a vectorized residual-permutation null. Listing 2 is one battery member: the same statistic in a family's linearizing coordinates with a wild bootstrap null.

Listing 1 — Method A wrapper: permutation curvature test (common.py, abridged).

```
def form_check_with_p(data, target, features, seed, b_perm=999):   # (abridged)
    # Method A wrapper: statistic = coefficient of the mean-square column
    # in log-log space; null = residual permutation, vectorized via pinv row.
    fc = dimreflex_form_check(data, target=target, features=list(features),
                              n_boot=200, random_state=seed)
    if not fc.get("checked", False):
        return {..., "fail": classify_reason(fc["reason"])}

    lx = np.log(data[list(features)].values)
    ly = np.log(data[target].values)
    n = len(ly)
    quad = (lx[:, 0] - lx[:, 0].mean())**2 if len(features) == 1 \
           else ((lx - lx.mean(axis=0))**2).mean(axis=1)
    base = np.column_stack([np.ones(n), lx])
    design = np.column_stack([base, quad])

    coef, *_ = _ols(design, ly)
    c2 = float(coef[-1])                    # statistic: curvature c2
    if not np.isclose(c2, fc["curvature"], rtol=1e-6, atol=1e-10):
        return {..., "fail": "statistic_mismatch"}   # native-drift sentinel

    bcoef, *_ = _ols(base, ly)              # restricted fit -> perm null
    pred, resid = base @ bcoef, ly - base @ bcoef
    if float(resid @ resid) <= 1e-12 * float(ly @ ly):
        return {..., "fail": "degenerate_zero_residual"}

    rng = np.random.default_rng(int(seed) * 1_000_003 + 17)
    pinv_row = np.linalg.pinv(design)[-1, :]     # c2 is linear in y
    c2_base = float(pinv_row @ pred)
    idx = np.argsort(rng.random((b_perm, n)), axis=1)
    null_c2 = c2_base + resid[idx] @ pinv_row    # B permutations at once
    mu, sd = float(null_c2.mean()), float(null_c2.std())
    p_emp = (1.0 + float(np.sum(np.abs(null_c2 - mu)
                                >= abs(c2 - mu)))) / (b_perm + 1.0)
    return {"p": float(p_emp), "rej": bool(p_emp < ALPHA),
            "t": float((c2 - mu) / sd)}
```

Listing 2 — Battery member: family-coordinate curvature test with wild bootstrap null (battery.py, abridged).

```
FAMILIES = {                                    # (u(x), v(y)) per family
    "power":       (np.log,          np.log),
    "exponential": (lambda x: x,     np.log),
    "logarithmic": (np.log,          lambda y: y),
    "rational":    (lambda x: 1.0/x, lambda y: 1.0/y),
}

def spec_test_wild(X, y, uf, vf, seed, b_wild=999):   # one battery member
    # H0: conditional mean linear in family coordinates (u, v).
    # Statistic: coefficient of quad = mean_j(U_j^2); null: wild bootstrap.
    U = _standardize_cols(uf(X))         # per-column transform + standardize
    v = vf(y)
    if (X <= 0).any() or (y <= 0).any():
        return {"p": np.nan, "fail": "nonpositive"}
    base = np.column_stack([np.ones(len(v)), U])
    quad = (U**2).mean(axis=1)
    design = np.column_stack([base, quad])
    coef, *_ = np.linalg.lstsq(design, v, rcond=None)
    t_obs = float(coef[-1])
    b, *_ = np.linalg.lstsq(base, v, rcond=None)
    resid = v - base @ b
    if float(resid @ resid) <= 1e-12 * float(v @ v):
        return {"p": np.nan, "fail": "degenerate_zero_residual"}

    rng = np.random.default_rng(seed)
    pinv_row = np.linalg.pinv(design)[-1, :]     # c2 is linear in v
    c2_base = float(pinv_row @ (base @ b))
    signs = rng.choice((-1.0, 1.0), size=(b_wild, len(v)))
    null = c2_base + (resid[None, :] * signs) @ pinv_row   # Rademacher wild
    mu, sd = float(null.mean()), float(null.std())
    p = (1.0 + float(np.sum(np.abs(null - mu) >= abs(t_obs - mu)))
         ) / (b_wild + 1.0)
    return {"p": float(p), "t": float((t_obs - mu) / sd)}
```

Appendix G — Change notes (v2 → v5)

1. **Suite S5 added (gate comparison).** The question "why not just compare against a nested rival?" is promoted from an anticipated objection to a measured result: AICc-quad leads all four cells of Table 12; the permutation gate is repositioned from best-calibrated test to self-calibrating portfolio member. The repositioning is reflected in the abstract ("Fifth"), Q6, §8, and the title's framing ("gates" rather than "tests").

2. **Suite S6 added (real data).** World Bank macro relations with a preregistered list and a preregistered dilution probe; unanimous strong-signal cells and mechanism-consistent divergence, including a live reproduction of the DILUTION signature (Table 14). No rate claims.

3. **Totals updated.** 1,345 → 1,480 cells; 488,260 → 508,825 fitted datasets.

4. **S3 raw table dropped from body** (Table 5 in v2) to keep the table count within workshop limits; the raw family map remains in released per-cell data and the calibrated Table 5 carries the claim. v2's Q2 and the "external validity" sentence in Q2 are superseded by the three-level dilution result (constructed → routed → observed).

5. **References extended** [12–14] (AICc, CV survey, World Bank data).

6. **Post-v3 addition: Appendix E (real-engineering bench).** A nine-source engineering companion to S6 — 91 datasets, 150 preregistered relations, 761,515 rows — run as a single-realization smoke pass whose cells stand outside the seeded totals (unchanged): unanimous detection of a documented rating-curve misspecification, echoes of the small-sample power gap and the interaction-by-second-feature structure, all three registered HC3 failure modes, and mutual misses on locally approximate laws.

7. **Major revision (v4) — reviewer response.** (a) The OR(A, E) combination's recall advantage is re-analyzed at the combination level: its naive joint false-positive budget is .065–.102, and recalibrated to a 5% joint budget it matches but never beats F (§4.7(d)); "best cell" claims withdrawn. (b) Fit-count Wilson intervals are supplemented with equation-level bootstrap intervals, leave-one-equation-out ranges, and per-equation rates (§4.1, Appendix C); the HC3 failure is shown to concentrate in 5 of 13 equations. (c) Permutation-null claims conditionalized throughout: residual permutation is an approximation calibrated empirically on our pools, not exact by construction (§3.2, §6). (d) **The HC3 claim is retracted and re-attributed**: an independent statsmodels check shows our pinv-based HC3 path — not the HC3 correction — produces the 42–56% false alarms (1.00 vs .067 on the worst tasks; ≥3-digit agreement elsewhere); Tables 2/3/6 C/D rows are now labeled path-specific (Q3). (e) S5 calibration is validated out-of-sample by leave-one-equation-out threshold transfer (§4.7(c)). (f) Real-data claims downgraded to signature-consistency with confounds stated (§3.6, §4.8, Appendix E); S6's five seeds reframed as Monte Carlo replicates with measured spread. (g) Battery set-level errors and multiplicity reported (§4.6). (h) Related work extended to the wider specification-testing literature; references [15]–[22] split with access dates and per-source licenses; [8] description corrected; Table numbering restored to consecutive (v3 left a gap at 5).

8. **Second-round revision.** (a) Independent verification extended to exog-HC3 (D): all 13 H0 equations, stratified sampling, three-path comparison — our SVD path matches statsmodels ($|\Delta p| \leq .0023$, covariance $\leq 3\times10^{-4}$) while the $X'X$ path fails only at measured $\text{cond}(X'X) \geq 4\times10^{17}$; the numerics paragraph now states the exact call (NumPy 2.5.2 pinv, rcond = 10^{-15} relative) and replaces informal precision arithmetic with measured condition numbers (Q3). (b) The .067 re-verification restated as 8/120, Wilson [.034,.126] — "compatible with nominal", not precision calibration (Q3, §4.1). (c) Task-level paired bootstrap added for calibrated power: Δ(D−A), Δ(D−B), Δ(D−C) intervals with win fractions; "D strongest" softened to "highest average power in the current task pool" (§4.2, Appendix C). (d) Glyph coverage audited across the compiled PDF (zero uncovered characters; the reported missing-symbol boxes could not be reproduced). (e) S2 counts unchanged (32 tasks: 13 H0 + 19 H1; no task was removed; 1,480 cells and 508,825 datasets stand as audited).

9. **Goal-switch revision (post-v4): the entailment claim promoted to the paper's thesis.** The title, abstract, introduction, and conclusion now center the claim the evidence was built to test: fit quality and functional correctness are not connected by necessity, in either direction. New Section 4.10 (Table 15) quantifies both directions on the quality benches (57,000 synthetic + 240,000 real-formula fits, C/D on the SVD path): within $R^2 \geq 0.99$, true forms extrapolate at 0.1–0.2% error while wrong forms sit at 100% >10% (synthetic) and 9.7% (real); wrong forms below $R^2 = 0.5$ are jointly missed by all four gates at 57.5% (n=50), and ii.11.28 ($R^2 \approx 0.002$) at 83%. New Q9 states the mechanism: detection tracks direction alignment corr($\hat{y}^2$, residual), not fit magnitude. The gates are repositioned throughout as measuring instruments for the entailment question, not as competing detectors; all prior suite results, retractions, and audit chains stand unchanged. The analysis is regenerated by `no_causal_link.py` (twelve anchor checks, build-failing on drift).

10. **Review round 3 (six-item): bounded claims, joint release, SVD everywhere, conflict fixes.** (a) The thesis is re-bounded per review: "fit quality cannot by itself certify functional form; the evaluated specification gates remain jointly blind to some form mismatches" — the stronger "no entailment / no information / logical independence" wording is withdrawn (title, abstract, introduction, Q9, conclusion), and the analysis script's name is superseded by `svd_main_results.py` reporting the joint-release accounting. (b) The headline quantity is now the *joint event on the same models* (Section 4.10, Table 15): dangerous wrong forms (extrapolation error >10%) passing the $R^2 \geq 0.99$ screen and jointly missed by all four gates — 0.67% of screened wrong forms, 6.9% of dangerous ones, release rate of dangerous forms 7.8% → 0.2% with n; certificate LR reported (dangerous wrong forms are 65× less likely to hold a full-green certificate than true forms — the certificate is informative but non-excluding). Low-fit silences (ii.11.28) are repositioned as mechanism, not epidemiology — a model at $R^2 \approx 0.002$ would not survive any sane fit screen. (c) All main-text C/D method comparisons now run on the verified SVD path (Tables 2, 3, 5, 15); the pinv(X′X) numbers survive only in Q3 as the numerical-implementation case study. This changes results substantively: D (exog-HC3, SVD) has the highest calibrated pooled power at every n on real formulas (cross-equation bootstrap confirming at $n \geq 200$); the previous draft's "A leads at small and medium n" was an artifact of thresholds derived from artifact-contaminated p-mass. Table 5's trigonometric C/D entries similarly un-suppressed (.58/.86 under SVD). (d) Conflicts fixed: §4.6 empty-set wording now states the two-pool semantics (V never empty on member-truth cells; V = Ø is the correct outcome on out-of-family cells, at Table 10's rates); Table 12 excludes the OR row (own-5% budget ≠ per-gate 5%) and reports it separately; §4.7(d) marks synth n=50 OR as uncalibratable (A alone fires at .080); the 805,825 total is replaced by the split accounting (508,825 fitted + 297,000 re-evaluated). (e) Direction geometry is offered as a diagnostic, not an identified mechanism (circularity acknowledged; the independent experiment — vary alignment at fixed misfit — is named as future work).

9. **Third-round revision (reviewer response).** (a) **Denominators are explicit everywhere.** Table 15 previously printed C/B in its last column ("1.5% of screened / 13.2% of dangerous") while §4.10(b) and the abstract reported C/D ("7.8% → 0.2%") under the same label; the two ratios are now separate columns with their denominators named, and the n = 50 divergence (350 released of 2,655 screened-dangerous, against 966 gate-silent of 12,369 all-dangerous) is stated in the table note. (b) **The n-gradient claim is weakened.** §4.10(b) no longer reads "sample size closes the release channel": it reports both denominators, states that at $n \geq 200$ every released dangerous model had already passed the screen, and names the three mixed effects (rising gate power, a changing numerator and denominator, the contracting best L_2 approximation within the power-law class), with the decomposition left outstanding. (c) **Threshold conventions flagged.** The 10% danger bound and the $R^2 \geq 0.99$ screen are labelled conventions in §4.10, and the screen-versus-gate budget asymmetry — an absolute goodness screen against a significance-level gate — is stated rather than left implicit. (d) **Certificate likelihood ratio corrected** to 65× (.845/.013; the earlier 50× came from rounding LR ≈ .02), in the abstract, §1, §4.10(c), §5, §8 and here. (e)

Unlike quantities are no longer juxtaposed: the synthetic bench's 100% is labelled a design property of that bench, and 9.7% is named as the dangerous share of *screened* wrong forms. (f)

Appendix D is regenerated on the SVD path. The previous draft's captions had drifted from the published images (D.1 was a CDF panel, not the retracted 42–56% claim; D.2 was the *uncalibrated* map; D.3 plotted S1 scenarios; D.4's ROC carried the pinv artifact), so the captions were first corrected to match the images and the panels were then re-rendered from the released cells

(`paper/review3_analysis.py`, §3.7), with §7's bit-identical claim scoped to table cells and §6 recording the provenance. (g) **Appendix C re-derived on the SVD path** (C .053 [.051–.056], D .056 [.054–.059]; task-level C [.045, .068]; leave-one-out [.048, .062]), with the pinv figures kept as the path-contrast case; the §4.1 sentence that attributed a pinv interval to the SVD path is corrected. (h) **Table 12's "matched FPR" claim scoped**: it holds in three of four columns; synth n = 50 is reported as measured because A fires at .080 there, the same standard the OR row is held to. (i) **§4.3 sensitivity readings downgraded** to directional — 30 seeds per cell cannot carry a 15-point trigger. (j) **The SR-method gap is stated as a gap** (Section 6): no symbolic-regression method is run anywhere in the benchmark, so the measured object is gate behavior on a pre-specified power-law candidate, not the pass rate of method-emitted wrong forms; that experiment is named as the next suite.

7. **Suite S7 added (joint-silence rate).** The panel's silence is now measured with the fit screen removed, on an independent re-sampling of S2's 18 non-monomial equations (2,000 datasets per task per n; 36,000 per n). Joint silence — no gate rejects and no gate declines — is 37.8% / 29.8% / 23.2% naive and 38.6% / 30.2% / 23.6% H0-calibrated at n = 50 / 200 / 1000 (Table 16). This is a different quantity from §4.10's pipeline yield and the two are never pooled: §4.10 conditions on the $R^2 \geq 0.99$ screen, S7 does not. Against the synthetic bench's 0.15–0.22% on the same unscreened basis the gap is 170–250×. Adding the S5 model-comparison gates at matched calibration moves the n = 50 figure from 38.6% to 42.2%, against 90.5% silence on true power laws (LR ≈ 2.2×; posterior ≈ 32% under a 50/50 prior). **Provenance:** `runs/joint_silence/`; every number is regenerated from `joint_silence_srsd_per_task.csv`, whose values disagree with that directory's `REPORT.md` summary table (32.0% vs 37.8%; 70% vs 90.5% for the H0 baseline — annotated in-file, October 2026); the summary is not used anywhere, and no Wilson intervals are given because equations, not datasets, are the independent unit. Abstract, §4's table index, §6, and §8 are updated accordingly.